The construction of four-dimensional quantum Yang–Mills theory

Under construction. Version of November 11, 2026: this manuscript is incomplete. Sections are added and revised as they are written and checked against the formal proof; see the status table at https://github.com/scottnarmstrong/yang-mills-informalization/tree/main/manuscript .

Abstract

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Part I
The theorem and the architecture of its proof

1 Introduction

2 Wilson theory

The starting measure assigns a group element to each lattice link and weights the curvature around each plaquette. We first fix the matrix normalization: it determines both the Wilson energy and the coefficient of the logarithmic coupling flow.

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, coordinate indices run from 1 to 4, and e_a is the ath coordinate vector. Time is the fourth coordinate. For the later relativistic statements write

$$\mathsf{m}(x) = (x^4)^2 - |\vec{x}|^2, \qquad \overline{V}_+ = \{\mathbf{p} : |\vec{\mathbf{p}}| \le \mathbf{p}^4\}\,.$$

Inner products are antilinear in the first variable, and spacetime integrals use Lebesgue measure. Arity zero is included; empty sums are zero and empty products are one. The group H_4 consists of the signed permutation matrices; $\mathsf{s}_a^{\mathrm{refl}}$ denotes reflection in coordinate a.

Definition 2.1 (Gauge normalization)**.** Fix $n_G \ge 1$ and a closed subgroup $G \subset U(n_G)$ with simple nonabelian real Lie algebra

$$\mathfrak{g} = \{X \in M_{n_G}(\mathbb{C}) : \exp(tX) \in G \text{ for every } t \in \mathbb{R}\}, \qquad \langle X, Y\rangle_{\mathfrak{g}} = -n_G^{-1}\operatorname{Re}\operatorname{tr}(XY)\,.$$

For an orthonormal basis $(e_\alpha^{\mathfrak{g}})$, put

$$d_{\mathfrak{g}} = \dim_{\mathbb{R}} \mathfrak{g}, \qquad C_{\mathfrak{g}} = -d_{\mathfrak{g}}^{-1}\sum_\alpha \operatorname{tr}_{\mathfrak{g}}\big(\operatorname{ad}(e_\alpha^{\mathfrak{g}})^2\big),$$

$$b_0 = \frac{11 C_{\mathfrak{g}}}{48\pi^2}, \qquad \kappa_G = 2b_0\,.$$

This fixes the normalization of the running coupling.

Here $\operatorname{ad}(X)Y = [X, Y]$ and $\operatorname{tr}_{\mathfrak{g}}$ is the real operator trace on $\mathfrak{g}$. Simplicity means $[\mathfrak{g}, \mathfrak{g}] \neq 0$ and no proper nonzero real ideal. Connectedness of G is not assumed. The adjoint maps are skew-adjoint for this metric; nonabelianity therefore gives $C_{\mathfrak{g}} > 0$ and $\kappa_G > 0$. For $u \in G$, set

$$\operatorname{pr}(u) = 1 - n_G^{-1}\operatorname{Re}\operatorname{tr} u, \qquad \operatorname{ch}(u) = n_G^{-1/2}\|u - 1\|_{\mathrm{F}}, \qquad \operatorname{ch}(u)^2 = 2\operatorname{pr}(u)\,. \tag{2.1}$$

The first quantity is the plaquette energy; the second measures its distance from the identity. Both are invariant under conjugation and inversion.

Definition 2.2 (Wilson measure)**.** For an integer $s \ge 1$ and $\beta \in \mathbb{R}$, put $\Lambda_s = (\mathbb{Z}/s\mathbb{Z})^4$. A configuration assigns $U(y, a) \in G$ to every positive link, with inverse on the reversed link. Define

$$U_{y;ab} = U(y, a)U(y + e_a, b)U(y + e_b, a)^{-1}U(y, b)^{-1},$$

$$E_s^{\mathrm{W}}(U) = \sum_{y\in\Lambda_s}\sum_{a<b}\operatorname{pr}(U_{y;ab}),$$

$$d\mu_{s,\beta}(U) = Z_{s,\beta}^{-1}e^{-\beta E_s^{\mathrm{W}}(U)}\prod_{y,a} dh(U(y, a))\,.$$

Here dh is probability Haar measure and $Z_{s,\beta}$ normalizes the measure. This is the lattice theory to be scaled.

The normalization is the positive finite integral

$$Z_{s,\beta} = \int e^{-\beta E_s^{\mathrm{W}}(U)} \prod_{y,a} dh(U(y,a)) \,.$$

Indeed, the integrand is continuous and strictly positive on the compact configuration space. A site gauge transformation $\gamma_{\mathrm{g}} : \Lambda_s \to G$ acts by

$$U^{\gamma_{\mathrm{g}}}(y,a) = \gamma_{\mathrm{g}}(y) U(y,a) \gamma_{\mathrm{g}}(y+e_a)^{-1} \,.$$

It changes each plaquette by conjugation at its base point, so the energy is gauge invariant. Invariance of probability Haar measure under left and right multiplication then gives gauge invariance of the Wilson measure. Lattice isometries transport links and invert reversed ones. The measure is invariant under these transformations as well; the finite-lattice identities are proved in Section 9.

A general Wilson family, indexed by $N, p \in \mathbb{N}$, consists of strictly increasing positive integer sequences (L_p) and (D_N), real couplings $\beta_{N,p}$, a label set $\mathcal{T}$, and complex insertions $I_{N,p}(v, y; U)$. The index N controls the spacing, and p controls the physical volume. Set

$$a_N = D_N^{-1}, \qquad \mu_{N,p} = \mu_{L_p D_N, \beta_{N,p}}, \qquad \hat{y} = a_N \big(r_{L_p D_N}(y^a)\big)_{a=1}^4 \,, \tag{2.2}$$

where $r_s(j)$ is the integer representative of $j \in \mathbb{Z}/s\mathbb{Z}$ in $(-s/2, s/2]$. Thus the physical torus has side L_p.

Definition 2.3 (Lattice moments)**.** For a label list $v \in \mathcal{T}^n$, sites $y \in \Lambda_{L_p D_N}^n$, and a test $f : (\mathbb{R}^4)^n \to \mathbb{C}$, define

$$M_{N,p}^{\mathrm{lat}}(v; y) = \int \prod_{i=1}^n I_{N,p}(v_i, y_i; U) \, d\mu_{N,p}(U),$$
$$C_{N,p}^{\mathrm{lat}}(v; f) = a_N^{4n} \sum_{y \in \Lambda_{L_p D_N}^n} f(\hat{y}) M_{N,p}^{\mathrm{lat}}(v; y) \,.$$

These are the point correlations and their physical Riemann sums.

The construction below chooses the spacings, couplings and local insertions together. In this normalization, Theorem 3.6 gives

$$\frac{\beta_{N,p}}{\log(1/a_N)} \longrightarrow \kappa_G \qquad (N \to \infty), \quad \text{uniformly in } p \,. \tag{2.3}$$

Thus κ_G is the logarithmic slope of the bare inverse coupling. At each fixed physical volume the ultraviolet limit will use the full sequence $N \to \infty$; only the subsequent infinite-volume limit requires a subsequence.

3 Renormalization

Blocking integrates one scale against a probability kernel. A curvature action, a vacuum scalar and two activity families describe the new density. The inductive step controls these coordinates and the coupling; the multiscale theorem chooses the bare coupling to reach one prescribed endpoint.

Fix $\mathsf{b} > 0$. Throughout we use the common choices of Sections 9–13, made once for this group realization and penalty in the order of Section A. Their existence is proved with Theorems 3.5 and 3.6. Constants may depend on these choices; further dependence is stated where it occurs.

3.1 Scales and the block map

Levels decrease under blocking. With odd ratios $\ell_m \geq 3$, set

$$D_0 = 1, \qquad D_{m+1} = \ell_m D_m, \qquad a_m = D_m^{-1}, \qquad s_m = \log D_m \,. \tag{3.1}$$

The level-m torus has side $L_p D_m$ in lattice units and side L_p in physical units. Starting at cutoff N, blocking successively reaches levels $N-1, \dots, 0$.

The stage J_m records the available derivative order and bounds. The stages satisfy $4 \leq J_m \leq J_{m+1} \leq J_m + 1$ and $J_m \to \infty$. Choose the reference endpoint $\bar{\beta}$ sufficiently large, as in row 7 of Section A; in particular $\bar{\beta} > 1$. Define

$$\widehat{\beta}_m = \bar{\beta} + \kappa_G s_m, \qquad M_m^{\mathrm{cut}} = \exp\big((\log\log \widehat{\beta}_m)^2\big), \qquad t_m = \widehat{\beta}_m^{-1/2} M_m^{\mathrm{cut}}, \tag{3.2}$$

$$B_m = \lceil (\log \widehat{\beta}_m)^2 \rceil, \qquad \epsilon_m^{\mathrm{reg}} = \widehat{\beta}_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}} \,. \tag{3.3}$$

The reference coupling $\widehat{\beta}_m$ tracks the leading running slope. The factor M_m^{cut} enlarges the Gaussian field scale; t_m is the threshold separating small and large plaquette fields. The radius B_m sets the scale of the support weights. The quantities $\epsilon_m^{\mathrm{reg}}$ and $\epsilon_m^{\mathrm{for}}$ bound the regular and forced activity sums, respectively. These thresholds depend on the reference coupling, so they remain fixed as the actual coupling varies.

The allowed coupling interval is

$$\mathfrak{B}_m = [\widehat{\beta}_m/2, 2\widehat{\beta}_m] \,.$$

At stage j, write $\mathfrak{q}_j$ for the derivative order, ϖ_j for the spatial power, σ_j for the support exponent and $\omega_j^{\mathrm{RG}} \geq 0$ for the coupling drift bound. Their choices and growth conditions are in Section A.

At a step $m+1 \to m$, the fine links U and coarse links V lie on $\Lambda_{L_p D_{m+1}}$ and $\Lambda_{L_p D_m}$. Put $j = J_{m+1}$ and $\ell = \ell_m$. Write $\mathbb{B}_R = \{-R, \dots, R\}^4$. In a patch of radius $(j+3)\ell$ about the fine site ℓy, let $h_{y,z}$ be the chosen frame at offset z. For a coarse edge $e = (y, a)$ the framed straight connector is

$$Y_{e,z} = h_{y,z} \left(\prod_{r=0}^{\ell-1} U(\ell y + z + r e_a, a) \right) h_{y+e_a,z}^{-1} \,. \tag{3.4}$$

This compares the fine transport between the two coarse endpoints in their chosen frames. The averaging weights and the predictor on its minimizing branch are

$$w_z^{\mathrm{av}} = \big(\mathrm{Unif}(\mathbb{B}_{(\ell-1)/2})\big)^{*j}(z), \qquad Q_e^{\mathrm{pred}} = \operatorname*{argmin}_{u \in G} \sum_z w_z^{\mathrm{av}} \operatorname{pr}(Y_{e,z} u^{-1}) \,. \tag{3.5}$$

Here Unif is uniform probability mass and $*j$ denotes j-fold convolution. This branch is used when the displayed minimum is below the fixed predictor cutoff; otherwise a tag selects $Y_{e,z}$ with mass w_z^{av}. Symmetric minimizing frames and the alternative axial frames are specified in Section 10, Definitions 10.1–10.2. The finite tag list α_{tag} specifies all frame and predictor choices. Their product mass satisfies

$$\lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) \geq 0, \qquad \sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) = 1 \,. \tag{3.6}$$

Definition 3.1 (Exact block map)**.** For fine links U and coarse links V, define

$$k_\beta^{(m)}(U, V) = \sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) \prod_{e \text{ coarse}} \frac{e^{-\mathsf{b}\beta \operatorname{pr}(V_e Q_e^{\mathrm{pred}}(U, \alpha_{\mathrm{tag}})^{-1})}}{\int_G e^{-\mathsf{b}\beta \operatorname{pr}(u)} \, dh(u)},$$

$$(\mathcal{B}_\beta^{(m)} f)(V) = \int k_\beta^{(m)}(U, V) f(U) \, dU \,.$$

Both product measures are probability Haar measures, and $\int k_\beta^{(m)}(U,V)\,dV = 1$. The map integrates one scale without changing the partition integral.

For fixed U and tags, right invariance of Haar measure makes every coarse-link factor integrate to one. Summing the tag masses proves the normalization. Consequently, for integrable f,

$$\int (\mathcal{B}_\beta^{(m)} f)(V)\,dV = \int f(U)\,dU\,. \tag{3.7}$$

The branch choices, measurability and symmetry properties of this kernel are proved in Section 10.

3.2 The density and its norms

At level m, write $\Lambda = \Lambda_{L_pD_m}$. Let $\mathrm{Flag}_m(U)$ be the sites incident to a plaquette whose chord exceeds t_m. A support is healthy if it avoids this set. A regular activity E_i^{reg} has a site support X_i and an anchor $o_i \in X_i$. A forced activity K_i^{for} has a support Y_i and a nonempty inventory $I_i \subset Y_i$ of flagged sites. Its eligibility event is

$$\mathcal{E}_i^{\mathrm{elig}} = \{U : Y_i \cap \mathrm{Flag}_m(U) = I_i\} \cap \mathcal{E}_i^{\mathrm{extra}}, \qquad K_i^{\mathrm{for}}(U) = 0 \quad (U \notin \mathcal{E}_i^{\mathrm{elig}})\,. \tag{3.8}$$

The first test specifies exactly which sites of Y_i are flagged. The extra event $\mathcal{E}_i^{\mathrm{extra}}$ is a fixed Borel gauge-invariant test supplied with that activity; it reads only links with both endpoints in Y_i and the flags in Y_i. It restricts which local configurations can contribute that forced term. The activity families are countable.

The path action separates the kinetic coefficient from the quadratic corrections generated by integration. For the fixed level data, write $c_{\mathrm{dir}} > 16c_{\mathrm{lin}} > 0$ for the clipping constants and $R_{J_m}^{\mathrm{rel}} > 0$ for the relaxation length. Distances use the unit-lattice sup metric. Set

$$\begin{aligned}
\mathrm{clip}_t(r) &= \begin{cases} c_{\mathrm{dir}} r^2/2, & r \le t, \\ c_{\mathrm{lin}} t r, & r > t, \end{cases} \\
\chi_{m,x}^{\mathrm{hlth}}(U) &= \mathbf{1}\Big\{ \mathrm{ch}(U_{y;ab}) \le t_m e^{\mathrm{dist}(x,y)/R_{J_m}^{\mathrm{rel}}} \text{ for every } y, a < b \Big\}\,.
\end{aligned} \tag{3.9}$$

The clip keeps a positive linear penalty beyond the small-field threshold, while $\chi_{m,x}^{\mathrm{hlth}}$ switches off a quadratic row when a plaquette violates the threshold prescribed by its distance from x. For each plaquette orientation $\mathfrak{p} = (a,b)$, let $\mathcal{F}_{\mathfrak{p}i}(U,x)$ be a curvature $\mathrm{proj}_{\mathfrak{g}}(U_\square - U_\square^{-1})/2$, transported along a specified finite path to x, and put

$$\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}}(U,x) = \sum_i r_{\mathfrak{p}i} \mathcal{F}_{\mathfrak{p}i}(U,x)\,.$$

The projection is orthogonal for the trace metric of Definition 2.1. Let $\mathfrak{a}_i^{(q)}(U,x)$, $q = 2,3,4$, denote the fixed real invariant q-linear expressions in transported curvatures, multiplied by the health indicator of their combined site supports. Then

$$\begin{aligned}
(P^{\mathrm{path}})_{N,m}^{\mathrm{raw}}(\beta,U) &= \beta \sum_{x,\mathfrak{p}} \Big[\mathrm{clip}_{t_m}(\mathrm{ch}(U_{x;\mathfrak{p}})) + \tfrac{1}{2}\chi_{m,x}^{\mathrm{hlth}}(U) |\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}}(U,x)|^2 \Big] \\
&\quad + \beta \sum_{x,i} (\mathfrak{a}_i^{(3)} + \mathfrak{a}_i^{(4)})(U,x) + \sum_{x,i} \mathfrak{a}_i^{(2)}(U,x), \\
P_{N,m}^{\mathrm{path}}(\beta,U) &= |H_4|^{-1} \sum_{h \in H_4} (P^{\mathrm{path}})_{N,m}^{\mathrm{raw}}(\beta, h^* U)\,.
\end{aligned} \tag{3.10}$$

Here $x \in \Lambda$ and $i \in \mathbb{N}$; h^*U is the link transport by the lattice isometry h. The path coefficients and multilinear expressions are independent of p. Their absolutely convergent series and the clipping restrictions are given in Section 11, Definitions 11.1–11.2 and C.1. In particular the action is affine in β; its last quadratic sum is independent of β.

For these fixed path actions and thresholds, a *representation* at level m is the datum

$$\rho = \big(\beta, v_{\mathrm{vac}}, (X_i, o_i, E_i^{\mathrm{reg}})_i, (Y_i, I_i, \mathcal{E}_i^{\mathrm{extra}}, K_i^{\mathrm{for}})_i\big)\,. \tag{3.11}$$

The scalar v_{vac} is its vacuum term, and $\beta(\rho) = \beta$ its running kinetic coefficient. These coordinates describe the integrated density below.

Definition 3.2 (Effective density)**.** For regular activities $(X_i, o_i, E_i^{\mathrm{reg}})$ and forced activities $(Y_i, I_i, K_i^{\mathrm{for}})$, define

$$\operatorname{dens}\rho(U) = e^{v_{\mathrm{vac}}|\Lambda| - P_{N,m}^{\mathrm{path}}(\beta,U) - \sum_i \mathbf{1}_{X_i \text{ healthy}} E_i^{\mathrm{reg}}(U)} \cdot \sum_{\substack{\mathcal{A}_{\mathrm{for}} \subset \mathbb{N} \text{ finite} \\ Y_i \ (i\in\mathcal{A}_{\mathrm{for}}) \text{ pairwise disjoint} \\ \bigcup_{i\in\mathcal{A}_{\mathrm{for}}} I_i = \mathrm{Flag}_m(U)}} \prod_{i\in\mathcal{A}_{\mathrm{for}}} K_i^{\mathrm{for}}(U)\,.$$

The sum covers every flagged site; each inventory is a nonempty subset of its support. This represents the density after integration.

If there are no flagged sites, only the empty family occurs and the forced sum equals one. When flags are present, their exact coverage and the disjointness of the supports assign each flagged site to precisely one selected forced term. Regular activities enter only on healthy supports. Their local regularity and cancellations, and the restrictions on forced activities, are specified in Section 12, Definitions 12.1–12.2.

For a link e, let e^- be its initial site. The left derivative $D^L_{\dot{\mathcal{A}}}$ differentiates the increments $U_e \mapsto e^{t\dot{\mathcal{A}}_e} U_e$, with $\dot{\mathcal{A}}_e \in \mathfrak{g}$ and norm from Definition 2.1.

Definition 3.3 (Local derivative seminorm)**.** For a regular activity anchored at o_i, set

$$\|E_i^{\mathrm{reg}}\|_m = \sup_{0\le q\le \mathfrak{q}_{J_m}} \sup_{U,\dot{\mathcal{A}}_1,\dots,\dot{\mathcal{A}}_q} \big| D^L_{\dot{\mathcal{A}}_1} \cdots D^L_{\dot{\mathcal{A}}_q} E_i^{\mathrm{reg}}(U) \big|\,.$$

The second supremum is over configurations healthy on X_i and directions satisfying

$$|(\dot{\mathcal{A}}_h)_e| \le t_m \big(1 + \operatorname{dist}(e^-, o_i)\big)^{\varpi_{J_m}}\,.$$

This measures ordered local derivatives in units of the permitted field size.

For a support X, let $\mathrm{cov}_{B_m}(X)$ be the minimum number of radius-B_m lattice balls with connected union covering X. The exponential weight below penalizes supports that require many such balls.

Definition 3.4 (Weighted activity sums)**.** For nonnegative majorants $e_i^{\mathrm{maj}} \ge \|E_i^{\mathrm{reg}}\|_m$ and $k_i^{\mathrm{maj}} \ge \sup_U |K_i^{\mathrm{for}}(U)|$, define

$$\mathcal{E}_m = \sup_x \sum_{o_i = x} e^{\sigma_{J_m} \mathrm{cov}_{B_m}(X_i)} e_i^{\mathrm{maj}},$$

$$\mathcal{K}_m = \sup_x \sum_{x\in Y_i} e^{\sigma_{J_m} \mathrm{cov}_{B_m}(Y_i)} k_i^{\mathrm{maj}}\,.$$

These sums measure regular and forced terms separately, with a penalty for large supports.

The chosen admissible classes $\mathcal{A}_{N,m,p}$, initial classes $\mathcal{I}_{N,p} \subset \mathcal{A}_{N,N,p}$, and adjacent-level relations

$$\mathcal{R}_{N,m,p} \subset \mathcal{A}_{N,m,p} \times \mathcal{A}_{N,m+1,p}$$

are constructed in Section 12, D.2. They retain the Wilson initial representations, the coarse representations produced by the inductive assembly, and the estimates needed to continue that assembly and perform endpoint tuning. For every member, D.1 supplies majorants with convergent weighted sums satisfying

$$\mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}, \qquad \mathcal{K}_m \le \epsilon_m^{\mathrm{for}}\,. \tag{3.12}$$

These inequalities are properties of the chosen classes; the classes also retain the restrictions imposed by the inductive assembly. The relation $\mathcal{R}_{N,m,p}$ records its permitted adjacent pairs. The density identity and drift bound below hold for the pairs produced by RG and for the trajectories chosen in MT.

Along a continuous curve, supports, anchors, inventories and extra tests are fixed, and the coupling and vacuum scalar are continuous. Differences of regular derivatives on healthy configurations and of forced values admit majorants whose weighted sums over all addresses tend to zero as the parameter approaches any point. The weights are those of Definition 3.4 at the level of the representation, including t_m in the derivative seminorm. Thus the input and output curves below use levels $m+1$ and m, respectively. This is the continuity of Definition 12.3 in Section 12: regular derivatives are measured on the healthy domain, and forced terms are compared in supremum norm on all configurations. No derivative of a flag indicator is taken.

3.3 The inductive step

Exact integration removes one scale. Extracting its marginal curvature-square term into the coupling leaves a bounded drift error and returns a density in the next admissible class.

Theorem 3.5 (One step)**.** *For every N, p, $m < N$, reals $u_0 \le u_1$ and continuous $\rho'_u \in \mathcal{A}_{N,m+1,p}$ with $\beta'_u = \beta(\rho'_u) \in \mathfrak{B}_{m+1}$ for all $u \in [u_0, u_1]$, there is a continuous $\rho_u \in \mathcal{A}_{N,m,p}$ with $(\rho_u, \rho'_u) \in \mathcal{R}_{N,m,p}$ and, for every $u \in [u_0, u_1]$ and coarse configuration V,*

$$\begin{aligned} \operatorname{dens} \rho_u(V) &= (\mathcal{B}^{(m)}_{\beta'_u} \operatorname{dens} \rho'_u)(V), \\ \beta(\rho_u) &= \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \le \omega^{\mathrm{RG}}_{J_{m+1}}\,. \end{aligned}$$

The proof occupies Sections 10–12, ending with D.3. Symmetric gauge fixing and the stationary background give the small-field integration; the large-field lower bound controls the forced terms. The Gaussian determinant and the Haar-coordinate Jacobian contribute to the one-loop action. Its quadratic curvature coefficient changes the inverse coupling by $-\kappa_G \log \ell_m$, up to the displayed error. Fourth-order Wilson matching fixes the remaining local terms, and the scalar coupling reset absorbs the residual kinetic contribution in Section 11. Section 12 assembles the activities with the bounds (3.12).

For the fixed construction, ω_j^{RG} depends only on the stage; it is independent of N, p, the interval and the curve. Since $\widehat{\beta}_m = \widehat{\beta}_{m+1} - \kappa_G \log \ell_m$, a step changes $\beta - \widehat{\beta}$ by e_u only. RG imposes a band condition only on the input coupling. The trajectories of MT below also have their output couplings in the appropriate bands. Section 12, D.3 supplies the continuous initial curve

$$\beta \longmapsto \rho^{\mathrm{W}}_\beta \in \mathcal{I}_{N,p}, \qquad \beta(\rho^{\mathrm{W}}_\beta) = \beta, \qquad \operatorname{dens} \rho^{\mathrm{W}}_\beta = e^{-\beta E^{\mathrm{W}}_{L_p D_N}} \quad (\beta \in \mathfrak{B}_N)\,, \tag{3.13}$$

whose vacuum scalar is zero. This is the base case for the induction.

3.4 Multiscale trajectories

A trajectory is a tuple $\rho_{N,p,m} \in \mathcal{A}_{N,m,p}$, $0 \le m \le N$, satisfying

$$\rho_{N,p,N} \in \mathcal{I}_{N,p}, \qquad (\rho_{N,p,m}, \rho_{N,p,m+1}) \in \mathcal{R}_{N,m,p} \quad (m < N)\,.$$

Write $\beta_{N,p,m} = \beta(\rho_{N,p,m})$; here m is the final index, whereas it is the middle index of $\mathcal{A}_{N,m,p}$ and $\mathcal{R}_{N,m,p}$. To select the bare coupling, start with the continuous Wilson curve across $\mathfrak{B}_N$ and apply RG. At each lower level the drift bound makes the output couplings straddle a smaller interval inside the next band; the intermediate value theorem gives a closed subinterval on which all couplings remain in that band. Repeating this restriction reaches $\bar\beta$ at level zero. This is the shooting argument. It chooses the couplings for all N, p before an increasing sequence of volume indices $\mathsf{p} : \mathbb{N} \to \mathbb{N}$ is prescribed.

Theorem 3.6 (Multiscale trajectory)**.** *There are bare couplings $\beta_{N,p}$ and trajectories $(\rho_{N,p,m})_{m=0}^{N}$, for all $N, p \in \mathbb{N}$, with $\rho_{N,p,N} = \rho^{\mathrm{W}}_{\beta_{N,p}}$ and $\beta_{N,p,N} = \beta_{N,p}$, satisfying*

$$\beta_{N,p,0} = \bar\beta, \qquad \beta_{N,p,m} \in \mathfrak{B}_m \quad (0 \le m \le N),$$
$$\operatorname{dens} \rho_{N,p,m} = \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} \operatorname{dens} \rho_{N,p,m+1} \quad (m < N)\,.$$

For every $\varepsilon > 0$, some $m_0(\varepsilon)$ satisfies

$$|\beta_{N,p,m} - \kappa_G s_m| \le \varepsilon s_m \qquad (N \ge m \ge m_0,\ p \in \mathbb{N})\,.$$

The proof is the shooting argument of Section 13, E.1, with E.1.1–E.1.2. Fix this family of trajectories for all subsequent constructions. The exact drift bound of RG holds on every adjacent pair of these trajectories, and all their integrals equal $Z_{L_p D_N, \beta_{N,p}}$ by (3.7) and the Wilson start. The path arrays remain independent of p.

The running law follows by summing the drift from the fixed endpoint:

$$|\beta_{N,p,m} - \widehat{\beta}_m| \le \sum_{i=0}^{m-1} \omega^{\mathrm{RG}}_{J_{i+1}}\,. \tag{3.14}$$

Row 5 of Section A bounds the sum by $(\kappa_G/4)s_m$, which also places the trajectory couplings in $\mathfrak{B}_m$. Its stronger bound $\omega^{\mathrm{RG}}_{J_{i+1}} / \log \ell_i \le J_{i+1}^{-2}$ makes the sum $o(s_m)$: split at a fixed level, then bound the tail by $\sup_{i \ge i_0} J_{i+1}^{-2}$ times s_m. Since $J_i \to \infty$ and $s_m \to \infty$, the finite initial sum and $\bar\beta$ disappear after division by s_m, uniformly in $N \ge m$ and p.

The same level choices give $\log \ell_m \to \infty$ and $\log \ell_m / s_m \to 0$. The physical periods L_p are strictly increasing multiples of four, and every positive integer divides L_p for infinitely many p.

Endpoint tuning varies the prescribed coupling at level zero. Section 13, E.1.3 proves that there is a width $w_{\mathrm{tun}} > 0$, independent of N, p, such that the unique bare coupling $\beta_{N,p}(s)$ within the specified recursion that reaches $\bar\beta + w_{\mathrm{tun}} s$ satisfies $\beta_{N,p}(0) = \beta_{N,p}$ and extends to a function $b_{N,p}$, holomorphic on $|z| < 1$, with

$$b_{N,p}(s) = \beta_{N,p}(s) \qquad (-1 < s < 1)\,. \tag{3.15}$$

The real endpoint parameter ranges over $s \in [-1, 1]$. This analyticity is used in the full ultraviolet limit; it does not assert a common analyticity width for all correlation functions.

3.5 Local fields

Local fields are normalized along the same trajectories. Their polynomial labels specify the continuum Taylor coefficients, while transfer to a fixed reference level determines the insertion at the fine cutoff.

The labels use radial, or Fock–Schwinger, gauge: the condition in the next definition sets the radial component of the connection to zero.

Definition 3.7 (Polynomial labels)**.** Let $\mathcal{J}$ consist of formal $\mathfrak{g}$-valued arrays $\mathcal{A}_a(x) = \sum_{k\ge 0} \mathcal{A}_{k,a}(x)$, with $\mathcal{A}_{k,a}$ homogeneous of degree $k+1$ and

$$\sum_a x^a \mathcal{A}_{k,a}(x) = 0, \qquad V_d = \mathbb{C}[\mathcal{J}]^{\mathrm{Ad}(G)}_{\mathrm{weight}\le d}\,.$$

Each coordinate of $\mathcal{A}_k$ has bidegree $(k+2,1)$: engineering weight and number of polynomial factors. Fix bihomogeneous bases, the supremum coefficient norm, conjugation and the induced H_4 action. These labels describe invariant curvature polynomials; curvature has weight two and a covariant derivative adds one.

Write $\mathbf{1}$ for the constant polynomial one. Section 9, A.1 constructs a local section $P_d^{\mathrm{loc}}[\mathbf{c}](y;U)$, linear in $\mathbf{c}\in V_d$ and polynomial in the real and imaginary parts of link matrix entries on a fixed finite stencil $\Sigma_d \subset \mathbb{Z}^4$. It is gauge invariant, commutes with complex conjugation, is H_4 covariant, and sends the polynomial unit to one.

Its Taylor normalization compares lattice holonomies with a smooth connection jet. For $\mathsf{j} = (\mathcal{A}_k)_k \in \mathcal{J}$, put $\mathcal{A}^{\le d} = \sum_{k+2\le d} \mathcal{A}_k$ and let $U(h_{\mathrm{lat}},\mathsf{j})$ be its test holonomies at spacing h_{lat}. On a positive link (y,a), with $y\in\mathbb{Z}^4$, these are defined by

$$U(h_{\mathrm{lat}},\mathsf{j})(y,a) = u(1), \qquad u'(t) = u(t) h_{\mathrm{lat}} \mathcal{A}_a^{\le d}(h_{\mathrm{lat}}(y+te_a)), \qquad u(0) = 1_G\,.$$

The jet dilation is $\mathrm{sc}_{h_{\mathrm{lat}}}\mathsf{j} = (h_{\mathrm{lat}}^{k+2}\mathcal{A}_k)_k$. For every $\mathbf{c}\in V_d$ and $0\le q\le d$, A.1 proves

$$\frac{d^q}{dh_{\mathrm{lat}}^q} P_d^{\mathrm{loc}}[\mathbf{c}](0;U(h_{\mathrm{lat}},\mathsf{j}))\Big|_{h_{\mathrm{lat}}=0} = \frac{d^q}{dh_{\mathrm{lat}}^q}\mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}}\mathsf{j})\Big|_{h_{\mathrm{lat}}=0}\,. \tag{3.16}$$

Thus a homogeneous label of engineering weight w specifies the coefficient at order h_{lat}^w in this expansion.

Choose the reference level $m_{\mathrm{ref}}(d)$ with $J_{m_{\mathrm{ref}}(d)} \ge d+3$ and $\mathfrak{q}_{J_m} \ge d+1$, $\varpi_{J_m} \ge d$ for $m \ge m_{\mathrm{ref}}(d)$, as in Section 13, E.2.1. For a basis vector $b_{d,i}$ of bidegree (w_i,k_i) define the factor-count scaling by

$$\mathsf{D}_t^{\mathrm{fac}} b_{d,i} = t^{k_i} b_{d,i}\,.$$

This rescales each polynomial factor in units of the small-field threshold. For $N \ge m_{\mathrm{ref}}(d)$ and terminal input $\mathbf{v}\in V_d$, the marked density at the origin is initialized and evolved by

$$H^{\mathrm{ins}}_{N,p,N;d}(\mathbf{v};U) = \mathrm{dens}\,\rho_{N,p,N}(U)\, P_d^{\mathrm{loc}}[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}}\mathbf{v}](0;U), \tag{3.17}$$

$$H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v}) = \ell_m^d \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} H^{\mathrm{ins}}_{N,p,m+1;d}(\mathbf{v}) \quad (m_{\mathrm{ref}}(d)\le m < N)\,. \tag{3.18}$$

For fixed N,p,d, the transfer construction chooses coefficient vectors $\mathbf{x}_m(\mathbf{v})$ and regular and forced remainders $R^{\mathrm{reg}}_{m,i}(\mathbf{v})$ and $Q^{\mathrm{for}}_{m,i}(\mathbf{v})$. All are linear in $\mathbf{v}$. Write $\mathbf{x}_m = x_{m,0}\mathbf{1} + \mathbf{x}_m^\circ$, and abbreviate

$\rho_{N,p,m}$ by ρ_m. The regular remainders have supports $Z_{m,i}$ containing the origin. Define their sum and the forced remainder by

$$\begin{aligned}\mathcal{R}^{\mathrm{reg}}_m(\mathbf{v};U) &= \sum_i \mathbf{1}_{Z_{m,i}\ \mathrm{healthy}} R^{\mathrm{reg}}_{m,i}(\mathbf{v};U),\\ \mathcal{R}^{\mathrm{for}}_m(\mathbf{v};U) &= e^{v_{\mathrm{vac}}|\Lambda| - P^{\mathrm{path}}_{N,m}(\beta(\rho_m),U) - \sum_i \mathbf{1}_{X_i\ \mathrm{healthy}} E^{\mathrm{reg}}_i(U)}\\ &\quad\cdot \sum_{\mathcal{A}_{\mathrm{for}}\ \mathrm{eligible}} \sum_{i\in\mathcal{A}_{\mathrm{for}}} Q^{\mathrm{for}}_{m,i}(\mathbf{v};U) \prod_{j\in\mathcal{A}_{\mathrm{for}}\setminus\{i\}} K^{\mathrm{for}}_j(U)\,. \end{aligned} \tag{3.19}$$

All unmarked quantities in this formula are taken from ρ_m, and eligibility is the disjointness and exact flag coverage in Definition 3.2. The marked density has the exact decomposition

$$\begin{aligned}H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v};U) = \mathrm{dens}\,\rho_m(U)\left[x_{m,0}(\mathbf{v}) + \mathbf{1}_{\Sigma_d\ \mathrm{healthy}} P^{\mathrm{loc}}_d[\mathsf{D}^{\mathrm{fac}}_{t_m^{-1}}\mathbf{x}^{\circ}_m(\mathbf{v})](0;U)\right]\\ + \mathrm{dens}\,\rho_m(U)\mathcal{R}^{\mathrm{reg}}_m(\mathbf{v};U) + \mathcal{R}^{\mathrm{for}}_m(\mathbf{v};U)\,.\end{aligned} \tag{3.20}$$

Here the stencil Σ_d is placed at the origin on the level-m torus. If $10\,\mathrm{dist}(x,0) < L_pD_m$ for every $x \in Z_{m,i}$, the regular remainder $R^{\mathrm{reg}}_{m,i}$ has zero radial-jet Taylor coefficients through order d; the forced remainders vanish outside their unmarked eligibility events. Their format and bounds are proved in Section 13, E.2.2–E.2.2a. We use the fixed decomposition selected there. It supplies the invertible terminal map of E.2.1:

$$\mathbf{x}_N(\mathbf{v}) = \mathbf{v}, \qquad \mathbf{x}_{m_{\mathrm{ref}}(d)}(\mathbf{v}) = \mathsf{Z}_{N,p,d}\mathbf{v}\,.$$

For $N < m_{\mathrm{ref}}(d)$ choose invertible maps to define the insertions; transfer normalization is required only at and above the reference level.

Definition 3.8 (Renormalized insertions)**.** Put $\mathcal{T} = \bigsqcup_{d\ge 0} V_d$ and define, at every cutoff,

$$I_{N,p}((d,\mathbf{c}),y;U) = O^{(d)}_{N,p}(\mathbf{c},y;U) = a_N^{-d} P^{\mathrm{loc}}_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}}\mathsf{Z}^{-1}_{N,p,d}\mathbf{c}](y;U)\,.$$

This fixes the insertion along the trajectory. Its distinguished unit coefficient is $\mathbf{u}_d = a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$, whose insertion equals one for $N \ge m_{\mathrm{ref}}(d)$. The scaling $\mathsf{D}^{\mathrm{fac}}_t$ counts factors, not engineering weight.

The inverse terminal map makes the transferred coefficient equal to the prescribed $\mathbf{c}$ at the reference level. Thus $\mathbf{c}$ is specified at the fixed physical scale $a_{m_{\mathrm{ref}}(d)}$, in that level's lattice units. The factor $\ell^d_m = (a_m/a_{m+1})^d$ uses the degree label of the whole space V_d; the terminal map allows mixing between its engineering weights. The factor a_N^{-d} converts the normalized lattice insertion to physical units.

The unit identity follows from $\mathbf{x}_m(a^d_N\mathbf{1}) = a^d_m\mathbf{1}$, proved with the transfer symmetries in Section 13, E.2.1a. This explains why the coefficient giving the constant insertion is $a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$, even though V_d also contains polynomials of lower engineering weight.

The bare couplings of MT and these insertions now specify one Wilson family. Each insertion is continuous, gauge invariant and local on its fixed stencil. MT at $m = N$ gives the volume-uniform asymptotic slope $\kappa_G > 0$. These are the lattice inputs to the continuum limits.

4 The continuum limit

We first remove the lattice spacing at each fixed volume, then let the volume increase along one refinement. Throughout, the Wilson family and insertions are those of Section 3; all couplings $\beta_{N,p}$ and insertion maps are chosen before the increasing sequence of volume indices is prescribed. Constants may depend on the fixed group, its realization, the penalty and the construction choices of Appendix A. Additional dependence is stated below.

For positions in $(\mathbb{R}^4)^n$, set

$$\mathrm{Conf}_n = \{x : x_i \neq x_j \text{ for } i \neq j\}, \qquad \mathrm{Gen}_n = \{x : x_i^a \neq x_j^a \text{ for } i \neq j,\ a = 1, \dots, 4\},$$
$$\mathrm{NoHit}_{L,n} = \{x : x_i - x_j \notin L\mathbb{Z}^4 \text{ for } i \neq j\}.$$

The last domain excludes collisions on a torus of physical side L. A separated test is a smooth function on $(\mathbb{R}^4)^n$ with compact closed support in Conf_n. It is p-admissible if its support lies in $((-L_p/2, L_p/2)^4)^n$. Write $\mathrm{sep}_\infty(x) = \min_{i \neq j} |x_i - x_j|_\infty$, with value $+\infty$ for $n \leq 1$.

For a finite list $\boldsymbol{l}$ of pairs $(i, a) \in \{1, \dots, n\} \times \{1, \dots, 4\}$, define

$$\Delta_{\boldsymbol{l}} = \prod_{(i,a) \in \boldsymbol{l}} a_N^{-1}(\tau_{i,a} - 1), \qquad \partial_{\boldsymbol{l}} = \prod_{(i,a) \in \boldsymbol{l}} \partial_{x_i^a}.$$

Here $\tau_{i,a}$ shifts slot i forward by one lattice link. These differences measure derivatives in physical units. Torus separation is the minimum, over distinct slots, of $\min_{z \in L_p\mathbb{Z}^4} |\hat{y}_i - \hat{y}_j + z|_\infty$, with value $+\infty$ for $n \leq 1$. An element $v_i = (d_i, c_i) \in \mathcal{T}$ retains both the polynomial label and its degree; we call it a token.

Marked transfers along the trajectories of Theorem 3.6 give bounds independent of the volume. They control compactness for each fixed list of fields and derivatives.

Proposition 4.1 (Uniform difference bounds)**.** *For every $v \in \mathcal{T}^n$, finite difference list $\boldsymbol{l}$ and $r > 0$, there are $B(v, \boldsymbol{l}, r) \geq 0$ and $N_0(v, \boldsymbol{l}, r)$ such that, for every $N \geq N_0$, every p and every lattice tuple of torus separation at least r,*

$$|\Delta_{\boldsymbol{l}} M_{N,p}^{\mathrm{lat}}(v; y)| \leq B.$$

The proof is in Section 13, E.2.2–E.2.4. Marked transfers control insertion coefficients and remainders. The chessboard estimate reflects an insertion into neighbouring blocks and bounds a mixed correlation by the product of the resulting reflected moments. Choosing an intermediate level whose physical blocks fit below the prescribed separation permits its use for the differentiated insertions. The reflected-moment constant in E.2.3 is chosen before that level, although its cutoff threshold may depend on it. No common bound or cutoff threshold is asserted over unbounded labels or difference orders.

The common endpoint can be varied analytically. More precisely, E.1.3 supplies a width $w_{\mathrm{tun}} > 0$, independent of N, p, such that for each $s \in [-1, 1]$ exactly one bare coupling $\beta_{N,p}(s)$ within the specified recursion reaches $\bar{\beta} + w_{\mathrm{tun}} s$ at level zero. It equals $\beta_{N,p}$ at $s = 0$ and extends holomorphically to $|s| < 1$.

This tuning gives convergence along the full ultraviolet sequence. The limiting kernel is periodic in each position and smooth away from periodic collisions, including configurations where positions share some coordinates.

Theorem 4.2 (Fixed-volume limit)**.** *For every p, n and $v \in \mathcal{T}^n$, there is $M_{p,v}^{\mathrm{UV}} \in C^\infty(\mathrm{NoHit}_{L_p,n})$ such that, for every finite $\boldsymbol{l}$ and every p-admissible separated test f,*

$$\lim_{N \to \infty} a_N^{4n} \sum_{y \in \Lambda_{L_p D_N}^n} f(\hat{y}) \Delta_{\boldsymbol{l}} M_{N,p}^{\mathrm{lat}}(v; y) = \int f(x) \partial_{\boldsymbol{l}} M_{p,v}^{\mathrm{UV}}(x)\, dx.$$

For $\boldsymbol{l} = \varnothing$, the left side is $\lim_N C^{\mathrm{lat}}_{N,p}(v; f)$. Thus the inner limits in Definition 4.3 below exist at every admissible volume. Admissibility places the test support inside $\mathrm{NoHit}_{L_p,n}$, so the displayed integrand is compactly supported and integrable.

The proof is in Section 13, E.3.1–E.3.2. At a fixed intermediate level, the vacuum scalar cancels between the correlation numerator and its partition function. Analytic tuning and the fixed-level transfer estimates make the normalized integrals of the transferred insertions Cauchy as N increases. For insertion sites obtained by repeated blocking from common finer sites, comparison with the original correlations has error bounded by a constant times $e^{-B_m/64}$. First choose the intermediate level m, then take both cutoffs large; the triangle inequality gives the Cauchy property for the original correlations. UB extends convergence from the dense set of these sites and identifies every derivative. E.3.2 also treats multilinear interpolants evaluated at any sequence $x_N \to x \in \mathrm{NoHit}_{L_p,n}$, including when the points cross a boundary of the chosen fundamental cell. Thresholds may depend on p.

The following definition keeps the order of limits explicit. Write $C^{\mathrm{UV}}_p(v; f)$ for a fixed-volume smeared limit.

Definition 4.3 (Ordered convergence)**.** A Wilson family converges to K along $P : \mathbb{N} \to \mathbb{N}$ if, for every n, v and separated test f, the test is eventually $P(k)$-admissible and there are numbers $C^{\mathrm{UV}}_p(v; f)$ with

$$C^{\mathrm{lat}}_{N,p}(v; f) \longrightarrow C^{\mathrm{UV}}_p(v; f) \quad (N \to \infty) \qquad \text{at every admissible } p,$$
$$C^{\mathrm{UV}}_{P(k)}(v; f) \longrightarrow \int f(x) K_n(v; x)\, dx\,.$$

Values at inadmissible volumes are unrestricted.

A table assigns to each degree list $(d_i)_{i=1}^n$ and $x \in \mathrm{Conf}_n$ a complex multilinear function $S_n(c; x)$ of $c_i \in V_{d_i}$. For $v_i = (d_i, c_i)$ write also $S_n(v; x)$, and for the word $w = (v_i(x_i))_{i=1}^n$ write $S(w)$. Conjugation and the signed permutation action are those of the polynomial labels; a unit slot carries the distinguished coefficient $\mathbf{u}_d = a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$ of Definition 3.8. The following conditions specify the Euclidean properties of the table to be constructed.

Definition 4.4 (E1–E3. Euclidean symmetries)**.** E1 is simultaneous permutation symmetry, $S_0 = 1$ and deletion of a slot carrying $\mathbf{u}_d$. E2 is

$$S_n(hc; hx) = S_n(c; x) \quad (h \in H_4), \qquad S_n(\bar{c}; x) = \overline{S_n(c; x)}\,.$$

E3 is translation invariance.

A word is positive-side for axis a when all its positions have positive ath coordinate. The product of words concatenates their lists, so reflection positivity tests a correlation with one reflected group of insertions.

Definition 4.5 (E4–E5. Reflection positivity)**.** E4 requires, for every axis a, positive-side separated words w_i and complex coefficients z_i,

$$\sum_{i,j} \bar{z}_i z_j S(\mathsf{s}^{\mathrm{refl}}_a w_i^* w_j) \geq 0\,.$$

The star reverses the word and applies conjugation to its labels; $\mathsf{s}^{\mathrm{refl}}_a$ reflects positions and acts on labels through H_4. E5 requires the same positivity for finite sums of words smeared against joint smooth compactly supported tests on positive-side separated configurations, and absolute integrability of every mixed pairing.

For a fixed label list, D_x^k denotes the kth Fréchet derivative in the positions, with the product Euclidean norm. The next bound controls approach to collision sets uniformly over translations and over configurations with large diameter.

Definition 4.6 (E6–E7. Euclidean regularity)**.** E6 is smoothness on Conf_n. E7 requires, for each n, d_i, c_i, k, constants $C(n,d,c,k) \geq 0$ and an integer $A(n,d,c,k) \geq 0$ such that

$$\|D_x^k S_n(c;x)\| \leq C r^{-A} \qquad (0 < r \leq 1,\ \mathrm{sep}_\infty(x) \geq r)\,.$$

One volume refinement will serve every field and all subsequent constructions.

Theorem 4.7 (Infinite-volume limit)**.** *For every strictly increasing* $\mathsf{p} : \mathbb{N} \to \mathbb{N}$*, there are a strictly increasing* $\mathsf{q} : \mathbb{N} \to \mathbb{N}$ *and a multilinear table* S *such that the Wilson family converges to* S *along* $P = \mathsf{p} \circ \mathsf{q}$ *in the sense of Definition 4.3. The table satisfies E1–E7.*

The proof is in Section 14, F.1–F.3. The extraction first gives volume limits on Gen_n; the corrected difference estimates of F.3 extend them smoothly to Conf_n (E6) and give the power bounds E7. Uniqueness identifies each extracted fixed-volume kernel with $M_{p,v}^{\mathrm{UV}}$, retaining UV's full inner limit at every admissible volume. Lattice reflection positivity for finite sums of smeared words passes to the limit, giving E5; E4 follows from E5 together with E1 and E6. The same extraction fixes the corrected coefficient maps of F.2, which adjust labels of different degrees so their correlations agree or differ by a positional derivative. Ordered convergence to S persists along every further volume refinement. No uniqueness of the infinite-volume limit is asserted; fix p, its refinement q and the resulting table S from now on.

Write $v(x) = (v_1(x_1), \dots, v_n(x_n))$ for a spectator word. Two polynomial labels represent the same observable when no separated correlation distinguishes them.

Definition 4.8 (Observable fields)**.** Set

$$\begin{aligned}
\mathcal{N}_d(S) &= \big\{ c \in V_d : S_{n+1}(c(x_0), v(x)) = 0 \\
&\qquad\qquad \text{for every } n, v \text{ and } x \in \mathrm{Conf}_{n+1} \big\}, \\
F_d &= V_d / \mathcal{N}_d(S), \qquad \overline{[c]} = [\bar{c}], \qquad h[c] = [hc] \quad (h \in H_4), \\
\iota_d[c](v;x) &= S_{n+1}(c(x_0), v(x)), \qquad \mathcal{L} = \mathrm{span} \bigcup_d \iota_d F_d\,.
\end{aligned}$$

The maps ι_d give the responses to all spectators. Choose conjugation-invariant quotient norms.

E2 makes $\mathcal{N}_d(S)$ invariant under conjugation and H_4, so both displayed actions on F_d are well defined. Permutation symmetry makes correlations independent of representatives in every slot: for $\boldsymbol{A} = ([c_1], \dots, [c_n])$, put $S_n(\boldsymbol{A};x) = S_n(c;x)$. The unit field is $\mathbf{1}_d = [\mathbf{u}_d]$, with the normalization of Definition 3.8.

To state the dimension of F_d, introduce the free comparison space.

Definition 4.9 (Free Maxwell comparison)**.** For $d_{\mathfrak{g}}$ independent colours, take

$$\mathbb{E} A_a^\alpha(x) A_b^\beta(y) = \delta_{\alpha\beta}\delta_{ab}\Gamma(x-y), \qquad \Gamma(x) = \frac{1}{4\pi^2|x|^2}, \qquad F_{ab} = \partial_a A_b - \partial_b A_a\,.$$

Let M_d^{Max} consist of complex $\mathrm{Ad}(G)$-invariant Wick polynomials in curvature jets of weight at most d, modulo antisymmetry, the linear Bianchi identities and $\sum_a \partial_a F_{ab} = 0$. Their separated Wick correlations are S^{Max}.

The constant one is the Maxwell unit, coefficient conjugation gives its real structure, and inclusions retain the same polynomial at a larger weight bound. Free field differentiation is $\partial_a^{\mathrm{Max}}(:P:) =: \partial_a P :$ and the natural rotation action is $R_d^{\mathrm{Max}}(h)(:P:) =: hP :$, where the jet action is determined by

$$hF_{ab}^{\alpha} = \sum_{i,j} h_{ia}h_{jb}F_{ij}^{\alpha}, \qquad h\partial_a = \sum_i h_{ia}\partial_i h \quad (h \in O(4)) .$$

These operations preserve the defining relations and give the structures used in the comparison below.

The quotient fields admit inclusions that preserve correlations and derivatives that act as positional derivatives. Their dimensions equal those of the Maxwell spaces.

Proposition 4.10 (Field calculus)**.** *For the table S of Theorem 4.7, there are complex linear maps $\jmath_{d,e} : F_d \to F_e$ for $d \le e$ and $\partial_a : F_d \to F_{d+1}$ for every d, a such that, for every $A \in F_d$,*

$$\iota_e \circ \jmath_{d,e} = \iota_d, \qquad \iota_{d+1}(\partial_a A) = \partial_{x_0^a}\iota_d(A),$$
$$\dim_{\mathbb{C}} F_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}} .$$

The proof is in Section 14, F.2, using the corrected coefficient maps chosen with S. Each ι_d is injective by Definition 4.8, so the inclusions are unique and injective, compose as $\jmath_{e,f}\jmath_{d,e} = \jmath_{d,f}$ and satisfy $\jmath_{d,d} = \mathrm{id}$. By E1 they carry $\mathbf{1}_d$ to $\mathbf{1}_e$. Their images are nested, hence $\mathcal{L} = \bigcup_d \iota_d F_d$. These inclusions are characterized by correlations; they are not defined by the polynomial embedding $V_d \subset V_e$, since (d, c) and (e, c) have different insertion normalizations.

Permutation symmetry gives, for every arity n, degree list (d_i), $\boldsymbol{A} \in \prod_{i=1}^n F_{d_i}$, $x \in \mathrm{Conf}_n$, slot i and axis a,

$$\partial_{x_i^a} S_n(\boldsymbol{A}; x) = S_n(A_1, \dots, \partial_a A_i, \dots, A_n; x) .$$

E2 and the injectivity of the response maps imply that inclusions and derivatives commute with conjugation, that inclusions commute with H_4, and that

$$h\partial_a = \sum_b h_{ba}\partial_b h \qquad (h \in H_4) .$$

5 Stress and rotations

We compare the fields of Section 4 with Maxwell fields at small distances, then normalize a conserved tensor by its shell fluxes. The volume refinement and the table S remain fixed throughout. Subsequences taken below concern only comparison radii.

The comparison permits the basis of local fields to vary with the radius. This accommodates mixing among fields in a fixed finite-dimensional space.

Definition 5.1 (Scaling frames)**.** A scaling sequence consists of $r_k > 0$, $r_k \to 0$, and complex linear isomorphisms

$$\mathsf{U}_{k,d}^{\mathrm{sc}} : M_d^{\mathrm{Max}} \longrightarrow F_d .$$

It is stable if, for every $D \in \mathbb{N}$, these maps eventually preserve conjugation, H_4, units and inclusions exactly in all degrees $d \le e \le D$. Scaling dimensions are included in the maps: an ordinary comparison insertion is $\mathsf{U}_{k,d}^{\mathrm{sc}} u$ at $r_k x$, with no further factor r_k^d.

The frames compare fields without identifying their null ideals.

After rescaling positions and choosing these frames, the interacting correlations approach their Maxwell counterparts. The first comparison measures the error by its integral on separated compact sets; one sequence works for every fixed list of fields.

Proposition 5.2 (Maxwell comparison)**.** *There is one stable scaling sequence such that, for every arity n, degree list (d_i) and $u_i \in M_{d_i}^{\mathrm{Max}}$,*

$$S_n(\mathsf{U}^{\mathrm{sc}}_{k,d_i} u_i; r_k \cdot) \longrightarrow S_n^{\mathrm{Max}}(u;\cdot) \quad \text{in } L^1_{\mathrm{loc}}(\mathrm{Conf}_n)\,.$$

The proof is in Section 15, G.2.1. The lattice comparison stops near the requested physical scale and compares the resulting correlations with the free Gaussian ones. Finitely many separating correlations make the comparison maps injective; the upper bound $\dim_{\mathbb{C}} F_d \le \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$ from Proposition 4.10 makes them isomorphisms. Finite-dimensional corrections impose the eventual equalities of Definition 5.1.

The integral norm retains uniformity over tests. Indeed, write $g_k(x)$ for the difference of the two smooth kernels in Proposition 5.2. For compact $\mathcal{K} \subset \operatorname{int}\mathcal{K}' \subset \mathcal{K}' \subset \mathrm{Conf}_n$, smooth approximation to $\overline{g_k}/|g_k|$ with a cutoff equal to one on $\mathcal{K}$ gives

$$\int_{\mathcal{K}} |g_k(x)|\,dx \le \sup_{\substack{f\in C_c^\infty,\ \operatorname{supp} f\subset\mathcal{K}'\\ \|f\|_\infty\le 1}} \left| \int f(x) g_k(x)\,dx \right| .$$

Conversely, the tested error on $\mathcal{K}$ is bounded by $\|f\|_\infty \int_{\mathcal{K}} |g_k|$. Thus local integral convergence is equivalent to convergence uniform over the sup-norm unit ball of tests on each fixed compact set. This comparison is used to construct and normalize the conserved current before the stress tensor is available.

To read translation and angular charge, integrate a conserved symmetric tensor through a shell surrounding just one insertion. Let $A \in F_d$, let $\boldsymbol{B}$ be a list of n spectator fields at x, and take $(x_0, x) \in \mathrm{Conf}_{n+1}$.

Definition 5.3 (Shell fluxes)**.** For $J_{ab} = J_{ba} \in F_4$ with $\sum_a \partial_a J_{ab} = 0$ and $K_{\mathrm{rot}} \in \mathfrak{so}(4)$, choose a real shell cutoff $\chi \in C_c^\infty(\mathbb{R}^4)$ equal to one near x_0 and supported away from spectators. Define

$$\mathcal{P}_b^J = -\sum_a \int \partial_a \chi(y) S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\,dy,$$
$$\mathcal{Y}^J_{K_{\mathrm{rot}}} = -\sum_{a,b} \int \partial_a \chi(y) (K_{\mathrm{rot}}(y-x_0))_b S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\,dy\,.$$

These integrals measure the flux without meeting an insertion point.

The fluxes are independent of the shell cutoff. For two choices, their difference vanishes near every insertion; integration by parts and Proposition 4.10 remove the divergence term. In the angular flux the additional term is $\sum_{a,b} (K_{\mathrm{rot}})_{ba} J_{ab} = 0$, since K_{rot} is skew-symmetric and J is symmetric.

A field is real when it is fixed by conjugation. A family $J_{ab} \in F_4$ is H_4-tensorial when

$$hJ_{ab} = \sum_{i,j} h_{ia} h_{jb} J_{ij} \qquad (h \in H_4)\,.$$

It is conserved when $\sum_a \partial_a J_{ab} = 0$ in F_5 for every b, and centered when every component has zero one-point function. Section 15, G.2.2 proves that the centered real symmetric H_4-tensorial conserved tensors of degree four form a real line. Along a refinement of the sequence in Proposition 5.2, positive multiples of the rescaled nonzero current approach the free stress line in the degree-four frames. This identifies the current needed for the shell laws.

For a nonzero member J of this line, G.2.3 constructs complex linear charge maps $\mathsf{p}_b : F_d \to F_{d+1}$ and $\mathsf{Y}_{K_{\mathrm{rot}}} : F_d \to F_d$ satisfying, for every degree, field, spectator list, separated configuration and allowed cutoff χ,

$$\mathcal{P}_b^J = S((\mathsf{p}_b A)(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^J = S((\mathsf{Y}_{K_{\mathrm{rot}}} A)(x_0), \boldsymbol{B}(x)) .$$

The angular map is real linear in K_{rot}.

Section 15, G.3.1 identifies these maps with differentiation and rotations. It constructs continuous complex linear representations R_d of $O(4)$, extending H_4, fixing units and commuting with conjugation and inclusions. For a real $c_\star \neq 0$ common to every degree, the generators and charge maps satisfy

$$\mathsf{L}_d(K_{\mathrm{rot}}) = \frac{d}{dt} R_d(e^{tK_{\mathrm{rot}}})\Big|_{t=0}, \qquad \mathsf{p}_b = c_\star \partial_b, \qquad \mathsf{Y}_{K_{\mathrm{rot}}} = c_\star \mathsf{L}_d(K_{\mathrm{rot}}) .$$

The proof controls exterior angular flux before extending the rotation identity to all separated configurations. The resulting table has full covariance: for every arity n, degree list (d_i), $\boldsymbol{A} \in \prod_i F_{d_i}$, $x \in \mathrm{Conf}_n$ and $h \in O(4)$, G.3.2 gives

$$S_n(R_{d_i}(h) A_i; hx) = S_n(\boldsymbol{A}; x) .$$

For these representations, $O(4)$-tensorial means $R_4(h) J_{ab} = \sum_{i,j} h_{ia} h_{jb} J_{ij}$ for every $h \in O(4)$, with the same index convention as the discrete action above.

The common constant $c_\star$ fixes the normalization of the stress tensor by its action on fields.

Theorem 5.4 (Stress tensor)**.** *There is a real symmetric $O(4)$-tensorial family $T_{ab} \in F_4$ with $\sum_a \partial_a T_{ab} = 0$ such that, for every degree, field, spectator list, separated configuration and shell cutoff of Definition 5.3, every b and every $K_{\mathrm{rot}} \in \mathfrak{so}(4)$,*

$$\mathcal{P}_b^T = \partial_{x_0^b} S(A(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^T = S((\mathsf{L}_d(K_{\mathrm{rot}}) A)(x_0), \boldsymbol{B}(x)) .$$

The proof is in Section 15, G.3.1–G.3.2. Divide the current J by $c_\star$ and average its tensor transform over $O(4)$ with probability Haar measure. Covariance and the transformation laws for derivatives and generators give the same shell actions for each transformed current; integrating these identities preserves both actions. Symmetry, reality and conservation also pass through this finite-dimensional integral. Fix this averaged normalized tensor for all subsequent uses. Adding a real multiple of $\delta_{ab} \mathbf{1}_4$ preserves its shell laws, the limit below and the centered trace of Definition 7.1; no uniqueness is needed.

Write $\mathsf{L}_d^{\mathrm{Max}}(K_{\mathrm{rot}}) = \frac{d}{dt} R_d^{\mathrm{Max}}(e^{tK_{\mathrm{rot}}})\big|_{t=0}$ for the free rotation generators. The free stress T^{Max} is normalized by the same shell laws, with $\partial_a^{\mathrm{Max}}$ and $\mathsf{L}_d^{\mathrm{Max}}$, in G.1. Their sign gives

$$T_{ab}^{\mathrm{Max}} = -\sum_{c,\alpha} : F_{ac}^\alpha F_{bc}^\alpha : + \frac{\delta_{ab}}{4} \sum_{i,j,\alpha} : F_{ij}^\alpha F_{ij}^\alpha : .$$

The final comparison includes any chosen stress insertions. For a free list $u_1, \dots, u_n$ with $u_i \in M_{d_i}^{\mathrm{Max}}$, a stress-slot choice is a subset $I \subset \{1, \dots, n\}$ with component pairs (a_i, b_i) for $i \in I$. Put

$$A_i^0 = \begin{cases} T_{a_i b_i}^{\mathrm{Max}}, & i \in I, \\ u_i, & i \notin I. \end{cases}$$

The shell laws fix the stress normalization, so its comparison insertion needs only the engineering factor r_k^4. In the next limit D_x^j differentiates the whole function $x \mapsto S_n(\boldsymbol{A}_k; r_k x)$, including the rescaled positions.

Proposition 5.5 (Differentiated short-distance comparison)**.** *There is one stable scaling sequence such that, for every fixed free list and stress-slot choice, every derivative order j and compact $\mathcal{K} \subset \mathrm{Conf}_n$, with $A_{k,i} = r_k^4 T_{a_i b_i}$ on I for the fixed tensor of Theorem 5.4 and $A_{k,i} = \mathsf{U}_{k,d_i}^{\mathrm{sc}} u_i$ elsewhere,*

$$D_x^j S_n(\boldsymbol{A}_k; r_k x) \longrightarrow D_x^j S_n^{\mathrm{Max}}(\boldsymbol{A}^0; x) \quad \textit{uniformly on } \mathcal{K}\,.$$

The proof is in Section 15, G.3.3. With $I = \varnothing$ and $j = 0$, this gives Proposition 5.2 in uniform form along the same sequence: an integral against f is bounded by $\|f\|_\infty$ times the volume of its compact support times the uniform error. The earlier comparison is needed first to construct the tensor. Along the sequence of Proposition 5.5, for every fixed d, a, K_{rot}, the transported infinitesimal actions satisfy the operator-norm limits

$$r_k (\mathsf{U}_{k,d+1}^{\mathrm{sc}})^{-1} \partial_a \mathsf{U}_{k,d}^{\mathrm{sc}} \longrightarrow \partial_a^{\mathrm{Max}},$$
$$(\mathsf{U}_{k,d}^{\mathrm{sc}})^{-1} \mathsf{L}_d(K_{\mathrm{rot}}) \mathsf{U}_{k,d}^{\mathrm{sc}} \longrightarrow \mathsf{L}_d^{\mathrm{Max}}(K_{\mathrm{rot}})\,.$$

6 Phases and reconstruction

Keep the table and quotient fields of Sections 4–5. Compression to zero energy separates the mixed correlations into phases. The slab estimate of H.1.1 makes word insertion bounded after a positive time translation. The coordinate reflection $\mathsf{s}_a^{\mathrm{refl}}$ acts on labels through H_4.

Definition 6.1 (OS space and compression)**.** For words at distinct points in $x^a > 0$, let $\mathcal{H}_a^{\mathrm{OS}}$ be the completion of their complex span modulo the null space of

$$\langle w, v \rangle_a = S(\mathsf{s}_a^{\mathrm{refl}} w^* v), \qquad e^{-tH_a^{\mathrm{OS}}}[v] = [\tau_{te_a} v] \quad (t \geq 0)\,.$$

Let Ω_a^{OS} be the empty-word vector and P_a^0 the projection onto $\ker H_a^{\mathrm{OS}}$. For $h_{\mathrm{sl}} > 0$ and w with times in $(0, h_{\mathrm{sl}})$, define, on positive-time words v,

$$A_w v = w\, \tau_{h_{\mathrm{sl}} e_4} v, \qquad M^0(w) = P_4^0 \mathsf{B}_{h_{\mathrm{sl}}}(w)|_{\ker H_4^{\mathrm{OS}}}\,,$$

where $\mathsf{B}_{h_{\mathrm{sl}}}(w)$ extends A_w boundedly on OS classes, by H.1.1.

Translate any separated word into a slab to define its compression; H.1.2 proves independence of width and placement. The compressed operators generate the compression algebra

$$\mathfrak{A}_0 = C^*(M^0(w) : w \text{ separated})\,.$$

Section 16, H.1.2 proves that this algebra is separable, commutative and unital, and that Ω_4^{OS} is cyclic and separating. Its vacuum state is

$$a \longmapsto \langle \Omega_4^{\mathrm{OS}}, a \Omega_4^{\mathrm{OS}} \rangle, \qquad S(w) = \langle \Omega_4^{\mathrm{OS}}, M^0(w) \Omega_4^{\mathrm{OS}} \rangle\,.$$

The second identity uses translation invariance. The measure below represents this state on the character space; the separating property gives full support. A one-dimensional mixed ground space would make $M^0(w) = S(w)\mathrm{Id}$ and leave a single character.

Characters separate the mixed correlations without changing which fields are detectable. Call a character ξ a phase and S^ξ its correlation table.

Theorem 6.2 (Phase decomposition)**.** *The algebra $\mathfrak{A}_0$ has a compact metric character space X with a full-support probability measure ν such that, for every separated word w,*

$$S^\xi(w) = \xi(M^0(w)), \qquad S(w) = \int_{\mathsf{X}} S^\xi(w)\, d\nu(\xi)\,.$$

Each phase satisfies E1–E7 and has exactly the null spaces $\mathcal{N}_d(S^\xi) = \mathcal{N}_d(S)$. The identities of FC, $O(4)$ covariance and ST hold with S^ξ in place of S, for the same maps and tensor T. For fixed w, $\xi \mapsto S^\xi(w)$ is continuous.

The proof is in Section 16, H.1.1–H.1.3. Two positive coordinate directions give opposite orders to the middle two of four widely separated words. Rotation covariance identifies the limits, and the positive translation semigroups converge to the ground projections; the resulting identity makes the compressed operators commute.

For each phase apply Definition 6.1 to S^ξ, obtaining $\mathcal{H}^{\mathrm{OS}}_{4,\xi}$, $H^{\mathrm{OS}}_{4,\xi}$ and Ω^{OS}_ξ. Fix ξ_* in the full-measure set with one-dimensional time-axis OS ground space supplied by H.1.3. This choice uses no trace condition. Theorem 6.15 will extend ground-space uniqueness to every phase.

Write $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{4n}; \mathbb{C})$ and $\mathcal{S} = \mathcal{S}_1$. The following definitions specify the relativistic theory.

Definition 6.3 (Relativistic fields)**.** Take a complex Hilbert space $\mathcal{H}$, a complex vector space $\mathcal{L}$ of labels, a linear subspace $\mathcal{D} \subset \mathcal{H}$ and $\Omega \in \mathcal{D}$. Fields $\Phi(\ell, f) : \mathcal{D} \to \mathcal{D}$ are linear operators, complex bilinear in $(\ell, f) \in \mathcal{L} \times \mathcal{S}$. Include an antilinear map $\ell \mapsto \ell^*$, strongly continuous unitary translations $\mathsf{T}(z)$, $z \in \mathbb{R}^4$, and a Borel projection-valued measure E, with finite diagonal measures $\mu_\psi(B) = \langle \psi, \mathsf{E}(B)\psi\rangle$ for Borel $B \subset \mathbb{R}^4$.

Definition 6.4 (Poincaré action)**.** Take a topological group $\mathcal{P}$ with topological group isomorphism $\kappa_{\mathrm{P}} : \mathcal{P} \to SO^+(1,3) \ltimes \mathbb{R}^4$, writing $\kappa_{\mathrm{P}}(\gamma) = (\Lambda_\gamma, z_\gamma)$. Here $SO^+(1,3)$ preserves m, has determinant one and preserves $\overline{V}_+$. Include a strongly continuous unitary representation U with $\mathsf{U}(\kappa_{\mathrm{P}}^{-1}(1,z)) = \mathsf{T}(z)$ and linear bijections $\rho_{\mathrm{lab}}(\gamma)$ of $\mathcal{L}$. Set $(\gamma \cdot f)(x) = f(\Lambda_\gamma^{-1}(x - z_\gamma))$.

In group arguments we identify $\mathcal{P}$ with $SO^+(1,3) \ltimes \mathbb{R}^4$ through κ_{P}. The antilinear label map need not be involutive. The label action has no additional continuity, dimension or translation restriction. The spectral relation is imposed next.

Definition 6.5 (W0. Spectrum and vacuum)**.** For every $\psi \in \mathcal{H}$ and $z \in \mathbb{R}^4$, require

$$\langle \psi, \mathsf{T}(z)\psi\rangle = \int e^{i\mathbf{p}\cdot z}\, d\mu_\psi(\mathbf{p}), \qquad \mathsf{E}(\mathbb{R}^4 \setminus \overline{V}_+) = 0\,,$$

with Euclidean pairing. Every vector fixed by all translations belongs to $\mathbb{C}\Omega$.

Definition 6.6 (W1. Word domain and adjoints)**.** Require $\|\Omega\| = 1$, density of $\mathcal{D}$, and

$$\mathcal{D} = \mathrm{span}_{\mathbb{C}}\{\Phi(\ell_1, f_1)\cdots\Phi(\ell_k, f_k)\Omega : k \geq 0\}\,.$$

For each $\ell \in \mathcal{L}$, $\psi \in \mathcal{D}$, the map $f \mapsto \Phi(\ell, f)\psi$ is continuous from $\mathcal{S}$ to $\mathcal{H}$ in norm. For $\psi, \chi \in \mathcal{D}$, require $\langle \Phi(\ell, f)\psi, \chi\rangle = \langle \psi, \Phi(\ell^*, \bar{f})\chi\rangle$.

Definition 6.7 (W2. Covariance)**.** Require $\rho_{\mathrm{lab}}(1) = 1$ and $\rho_{\mathrm{lab}}(\gamma\gamma') = \rho_{\mathrm{lab}}(\gamma)\rho_{\mathrm{lab}}(\gamma')$. For every $\gamma \in \mathcal{P}$, $\ell \in \mathcal{L}$, $f \in \mathcal{S}$ and $\psi \in \mathcal{D}$, require $\mathsf{U}(\gamma)\mathcal{D} \subset \mathcal{D}$, $\mathsf{U}(\gamma)\Omega = \Omega$, and

$$\mathsf{U}(\gamma)\Phi(\ell, f)\psi = \Phi(\rho_{\mathrm{lab}}(\gamma)\ell, \gamma \cdot f)\mathsf{U}(\gamma)\psi\,.$$

Definition 6.8 (W3. Locality)**.** For $f, h \in \mathcal{S}$ with $\mathsf{m}(x - y) < 0$ for every $x \in \operatorname{supp} f$, $y \in \operatorname{supp} h$, require $[\Phi(\ell, f), \Phi(\ell', h)]\psi = 0$ for all $\ell, \ell' \in \mathcal{L}$ and $\psi \in \mathcal{D}$.

Definition 6.9 (Q2: Wightman axioms)**.** A relativistic datum of Definitions 6.3–6.4 is a Wightman theory if it satisfies W0–W3.

Definition 6.10 (Faithful labels)**.** The labels are faithful if, for every nonzero $\ell \in \mathcal{L}$, some $f \in \mathcal{S}$ has $\Phi(\ell, f) \neq 0$.

Definition 6.11 (Bounded weak commutant)**.** A bounded operator B belongs to the weak commutant if, for every smeared field Φ and all $u, v \in \mathcal{D}$, one has $u \in \operatorname{Dom}\Phi^*$ and $\langle \Phi^* u, Bv\rangle = \langle u, B\Phi v\rangle$. Here Φ^* is the unbounded Hilbert-space adjoint.

Definition 6.12 (Hermitian scalar labels)**.** A label is Hermitian if $\ell^* = \ell$, and Lorentz scalar if $\rho_{\mathrm{lab}}(\Lambda, 0)\ell = \ell$ for every $\Lambda \in SO^+(1,3)$. These conditions give the adjoint and scalar covariance identities through W1 and W2.

For reconstruction use the response space $\mathcal{L}$ of Definition 4.8. Put $\mathbb{F} = \bigsqcup_d F_d$ and $\iota(d, A) = \iota_d A$.

Definition 6.13 (Boundary condition)**.** For a phase ξ, a relativistic datum $(\mathcal{H}_\xi, \mathcal{L}, \mathcal{D}_\xi, \Omega_\xi, \Phi_\xi, \dots)$ of Definitions 6.3–6.4, linear maps $\iota_d : F_d \to \mathcal{L}$, and functions F_n^ξ on $\mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n$, require, for every $n \in \mathbb{N}$, $\boldsymbol{A} \in \mathbb{F}^n$, real $h_1 < \cdots < h_n$ and $f_i \in \mathcal{S}$,

$$\lim_{r \downarrow 0} \int \mathsf{F}_n^\xi(\boldsymbol{A}; (x_i^4 + irh_i)_i, (\vec{x}_i)_i) \prod_i f_i(x_i)\, dx \\ = \langle \Omega_\xi, \Phi_\xi(\iota(A_1), f_1) \cdots \Phi_\xi(\iota(A_n), f_n)\Omega_\xi\rangle\,.$$

For this formula alone a nonintegrable function has integral zero.

This boundary condition does not assert positive-height integrability, joint-test convergence or complex spatial analyticity.

Definition 6.14 (Ordered-time continuation)**.** For a phase ξ, put $\mathcal{Z}_n = \{z \in \mathbb{C}^n : \operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n\}$. A continuation F_n^ξ is holomorphic there at each fixed real spatial configuration and satisfies

$$\mathsf{F}_n^\xi(\boldsymbol{A}; (it_i)_i, \vec{x}) = S_n^\xi(\boldsymbol{A}; (\vec{x}_i, t_i)_i) \qquad (t_1 < \cdots < t_n)\,.$$

Together with Definition 6.13, it connects Euclidean and real-time correlations.

For a theory at phase ξ in the next theorem, write $\mathcal{H}_\xi$, $\mathcal{D}_\xi$, Ω_ξ, Φ_ξ and T_ξ for its Hilbert space, word domain, vacuum, fields and translations. Define its energy generator by $\mathsf{T}_\xi(te_4) = e^{itH_\xi}$. Its OS comparison map means the bounded linear map

$$\mathsf{J}_\xi^{\mathrm{OS}} : \mathcal{H}_\xi \longrightarrow \mathcal{H}_{4,\xi}^{\mathrm{OS}}, \qquad \mathsf{J}_\xi^{\mathrm{OS}}\Omega_\xi = \Omega_\xi^{\mathrm{OS}}\,,$$

sending each smeared field-word vector to the norm limit of its heat-regularized positive-time OS vectors as the ordered heights tend to zero. Section 16, H.2.2 constructs this map and proves its uniqueness.

Each character yields relativistic fields on the common response labels. The continuation and boundary condition specify their relation to the phase correlations.

Theorem 6.15 (Reconstruction)**.** *For every $\xi \in \mathsf{X}$, there are a Wightman theory (Definition 6.9) on $\mathcal{L}$, faithful in the sense of Definition 6.10, and functions F_n^ξ satisfying Definitions 6.14 and 6.13. Its OS comparison map $\mathsf{J}_\xi^{\mathrm{OS}}$ is onto, and*

$$\ker H_\xi = \mathbb{C}\Omega_\xi, \qquad \ker H_{4,\xi}^{\mathrm{OS}} = \mathbb{C}\Omega_\xi^{\mathrm{OS}}\,.$$

Every bounded operator in the weak commutant of its fields (Definition 6.11) is scalar.

The proof is in Section 16, H.2.1–H.2.4. Slab bounds control the analytic boundary values. For these continuations, A low-energy bound uniform in ξ, continuity of the vacuum-subtracted heat pairings and full support of ν extend almost-everywhere ground-space uniqueness to every phase. Thus the earlier choice of ξ_* remains valid. H.2.1–H.2.4 also give joint Schwartz distributions, phase averages and stress continuation.

The reconstructed label maps are common to all phases. In H.2.1 the adjoint is induced by time reflection followed by conjugation:

$$(\iota_d A)^* = \iota_d\big(R_d(\mathsf{s}_4^{\mathrm{refl}})\overline{A}\big)\,.$$

The Lorentz label action is the continuation of R_d: spatial rotations retain their Euclidean action, and the boost generators are $-i$ times the corresponding Euclidean rotation generators. In particular, a real $O(4)$-invariant field gives a Hermitian Lorentz scalar label. Retain the theory at ξ_* and suppress its phase subscript in $\mathcal{H}, \mathcal{D}, \Omega, \Phi$ and the other relativistic data.

7 Trace and products

The trace is centered in the phase already selected in Section 6. Its short-distance logarithm will distinguish the resulting theory from finite Wick comparisons.

Definition 7.1 (Centered trace)**.** For the tensor T fixed in Theorem 5.4 and $r > 0$, put

$$\tau = \sum_{a=1}^{4} T_{aa}, \qquad e_{\mathrm{tr}} = S^{\xi_*}(\tau), \qquad \Theta = \tau - e_{\mathrm{tr}}\mathbf{1}_4,$$
$$G_\Theta(r) = S_2^{\xi_*}(\Theta, \Theta; -re_4/2, re_4/2), \qquad A_G = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

The one-point value is position independent. Centering removes the vacuum contribution in the selected phase.

The stress tensor is centered in the mixture, but a single-phase one-point value need not vanish. The next limit detects a nonzero centered trace and determines its power and logarithm.

Theorem 7.2 (Trace law)**.** *The selected field Θ is nonzero, real and $O(4)$ invariant; its label $\iota_4\Theta$ is Hermitian and Lorentz scalar (Definition 6.12). For every $f \in \mathcal{S}$,*

$$\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0\,.$$

The full-radius limit is

$$\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0\,.$$

The proof is in Section 17, I.1.1–I.1.3. To identify the amplitude, use the free scalar curvature square

$$\mathcal{O}_{F^2} = \frac{1}{4}\sum_{a,b,\alpha} :F^{\alpha}_{ab}F^{\alpha}_{ab}:, \qquad S_2^{\mathrm{Max}}(\mathcal{O}_{F^2},\mathcal{O}_{F^2};-e_4/2,e_4/2) = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

Here $\alpha = 1,\dots,d_{\mathfrak{g}}$ indexes an orthonormal colour basis. At running coupling β, the scalar trace coefficient is $4d_2/\beta$, with $d_2 = -b_0/2$. Since $\kappa_G = 2b_0$, its leading coefficient at radius r is $4d_2/(\kappa_G \log(1/r)) = -1/\log(1/r)$. Squaring it and using the free correlation gives A_G. The comparison uses adjustable stopping levels and fourth-order information beyond Proposition 5.2; its estimates are uniform in the phase, although TR is stated only at the already selected ξ_*. For every prescribed sequence of positive radii tending to zero, the comparison holds along a subsequence. The positive spectral representation below makes G_Θ real and nonincreasing; a sandwich argument gives the limit on that subsequence. The subsequence criterion then gives the full-radius limit.

The same proof identifies centering on the word domain:

$$\Phi(\iota_4\Theta, f) = \Phi(\iota_4\tau, f) - e_{\mathrm{tr}}\left(\int f\right)\mathrm{Id}_{\mathcal{H}}\,.$$

It also constructs the unique positive tempered-growth spectral measure μ_Θ, supported in $\overline{V}_+$, for which

$$d\mu_{\Phi(\iota_4\Theta,f)\Omega} = |\widehat{f}_+|^2\, d\mu_\Theta, \qquad G_\Theta(r) = \int e^{-r\mathbf{p}^4}\, d\mu_\Theta(\mathbf{p}) \quad (r>0)\,.$$

Here $\widehat{f}_+(\mathbf{p}) = \int e^{i\mathbf{p}\cdot x} f(x)\,dx$. The measure's Lorentz invariance and its high-energy limits against compactly supported continuous tests and Schwartz tests are stated in I.1.2–I.1.3.

To define the comparison class, fix $m_{\mathrm{sh}} \geq 0$, integers $j_h, k_E \geq 0$ and continuous real h_{mom} on $\mathbb{R}^3$ with $h_{\mathrm{mom}}(\lambda\mathbf{k}) = \lambda^{j_h} h_{\mathrm{mom}}(\mathbf{k})$ for $\lambda > 0$.

Definition 7.3 (Finite Wick profiles)**.** Put $\omega_{m_{\mathrm{sh}}}(\mathbf{k}) = \sqrt{m_{\mathrm{sh}}^2 + |\mathbf{k}|^2}$ and, for $r > 0$,

$$K_{m_{\mathrm{sh}},k_E,h_{\mathrm{mom}}}(r) = \int_{\mathbb{R}^3} \frac{\omega_{m_{\mathrm{sh}}}(\mathbf{k})^{k_E} h_{\mathrm{mom}}(\mathbf{k})}{2\omega_{m_{\mathrm{sh}}}(\mathbf{k})} e^{-r\omega_{m_{\mathrm{sh}}}(\mathbf{k})}\, d^3\mathbf{k}\,.$$

A finite Wick profile is a function

$$W_{\mathrm{Wick}}(r) = \sum_{i=1}^{N} c_i \prod_{j=1}^{n_i} K_{m_{ij},k_{ij},h_{ij}}(r) \quad (r>0), \qquad c_i \in \mathbb{C}\,,$$

with finitely many terms and factors of the preceding kind; empty products are one. A zero denominator at zero momentum gives integrand zero.

The momentum integrals converge absolutely for $r > 0$. For a free scalar field of mass m_{sh}, the time-axis two-point function is $(2\pi)^{-3}K_{m_{\mathrm{sh}},0,1}(r)$. Derivatives give momentum factors, and Wick contractions give finite sums of products at the same radius.

Such profiles cannot reproduce the factor $r^{-8}\log^{-2}(1/r)$ in Theorem 7.2.

Proposition 7.4 (Finite-profile exclusion)**.** *For every finite Wick profile W_{Wick} and every $\delta > 0$, there is $0 < r < \delta$ such that*

$$G_\Theta(r) \neq W_{\mathrm{Wick}}(r)\,.$$

The proof is in Section 17, I.2.1. The mass-shell kernels have expansions to every finite order in integer powers of r and nonnegative powers of $\log r$. Finite products and sums therefore give, to bounded error, a finite sum of monomials $r^a(\log r)^k$ with $a \in \mathbb{Z}$ and $k \in \mathbb{N}$. Multiplication by $r^8 \log^2(1/r)$ makes the terms with $a > -8$ vanish. The remaining nonzero leading term diverges, so any finite limit is zero. The positive limit in Theorem 7.2 therefore excludes equality on every interval $(0, \delta)$. Only that limit and its positivity enter this exclusion. I.2.2 identifies polynomial mass-shell comparisons with their Laplace profiles and applies I.2.1; I.2.3–I.2.4 give the Fock countertests. Their comparison classes are defined there.

The product expansion is stated for the Euclidean tables on their common quotient fields. Write $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$ for a state index (a phase or the mixture), with $S^{\mathrm{mix}} = S$, and retain degree tags by $\mathbb{F} = \bigsqcup_d F_d$, $\deg(d, A) = d$. The underlying prescription consists of the original fields and their correlation functionals. From it, Section 18, J.1.1 constructs finite channel sets Ch_j, real fields $O_{j,\lambda} \in F_j$ and, for every finite I, functions

$$c_{j,\lambda} : \mathbb{F}^I \times (\mathbb{R}^4)^I \times \mathbb{R}^4 \longrightarrow \mathbb{C} .$$

Fix these choices once, before spectators, states and truncations. The fields form complex bases of conjugation-invariant complements of F_{j-1} in F_j, with $F_{-1} = 0$. The degree-zero vector spans the unit line; it need not equal the unit. For finite cluster and spectator index sets $I_{\mathrm{cl}}, I_{\mathrm{sp}}$, take positions $x \in (\mathbb{R}^4)^{I_{\mathrm{cl}}}$, $y \in (\mathbb{R}^4)^{I_{\mathrm{sp}}}$ and expansion center $o \in \mathbb{R}^4$.

Definition 7.5 (Product remainder)**.** For finite index sets $I_{\mathrm{cl}}, I_{\mathrm{sp}}$ and $D \in \mathbb{N}$, cluster and spectator lists $\boldsymbol{A} \in \mathbb{F}^{I_{\mathrm{cl}}}$, $\boldsymbol{B} \in \mathbb{F}^{I_{\mathrm{sp}}}$, put $|\boldsymbol{A}|_{\deg} = \sum_i \deg A_i$ and

$$\mathcal{R}_D^\zeta = S^\zeta(\boldsymbol{A}, x; \boldsymbol{B}, y) - \sum_{j=0}^{D} \sum_{\lambda \in \mathrm{Ch}_j} c_{j,\lambda}(\boldsymbol{A}; x, o) S^\zeta(O_{j,\lambda}(o), \boldsymbol{B}(y)) .$$

Use a fixed enumeration of $I_{\mathrm{cl}} \sqcup I_{\mathrm{sp}}$ for the concatenated correlations. A pair (x, o) is an $(r, M_{\mathrm{geo}}, \eta)$-cluster if $|x_i - o| \le M_{\mathrm{geo}} r$ and $|x_i - x_{i'}| \ge \eta r$ for distinct slots. Spectators are δ_{sp}-clear if $|o - y_j| \ge \delta_{\mathrm{sp}}$ and $|y_j - y_{j'}| \ge \delta_{\mathrm{sp}}$ for distinct slots. The remainder measures the error through degree D.

If $M_{\mathrm{geo}} r < \delta_{\mathrm{sp}}$, the concatenated positions are separated: $|x_i - y_j| \ge \delta_{\mathrm{sp}} - M_{\mathrm{geo}} r > 0$. Thus the remainder is defined in the small-radius range used below.

One field expansion approximates a cluster in every phase and in the mixture. Its coefficients are independent of all spectators.

Theorem 7.6 (Operator product expansion)**.** *For every finite list $\boldsymbol{A}$, $M_{\mathrm{geo}} \ge 0$, $\eta, \varepsilon > 0$, $j \in \mathbb{N}$ and $\lambda \in \mathrm{Ch}_j$, some $r_0 > 0$ gives, for $0 < r < r_0$ and every $(r, M_{\mathrm{geo}}, \eta)$-cluster,*

$$|c_{j,\lambda}(\boldsymbol{A}; x, o)| \le r^{j - |\boldsymbol{A}|_{\deg} - \varepsilon} .$$

For every pair of finite lists $\boldsymbol{A}, \boldsymbol{B}$, $M_{\mathrm{geo}} \ge 0$, $\eta, \delta_{\mathrm{sp}}, \varepsilon > 0$ and $D \in \mathbb{N}$, some $r_1 > 0$ gives, for $0 < r < r_1$, such clusters, δ_{sp}-clear spectators and all states,

$$|\mathcal{R}_D^\zeta| \le r^{D+1-|\boldsymbol{A}|_{\deg} - \varepsilon} .$$

The proof is in Section 18, J.1.1–J.1.2. A fixed left inverse recovers $A \in F_j$ from finitely many values $S(A(0), B)$ against chosen probe words B. Apply it to cluster correlations, average under

rotations and conjugation, and recursively subtract lower degrees. The lattice product estimate bounds the discarded part in the compression algebra. Since $|\xi(a)| \le \|a\|$ for $a \in \mathfrak{A}_0$, that bound holds in every phase, and integration gives the mixture, with one radius threshold. The exponents compare output and input degrees. Since $Cr^{\varepsilon/2}\log^m(1/r) \to 0$ for fixed C, m, the loss ε absorbs logarithms and constants, giving coefficient one for smaller radii. The thresholds have separate dependencies.

J.1.3–J.1.6 give integrated and nested estimates; the nested error is a sum of gains from proper clusters.

Contact subtractions use the original fields and correlations underlying J.1.1, with separate projections extracting fields through finite correlation probes. Their clearance is a radius keeping the collapsed cluster separated from these probes, independent of spectator clearance δ_{sp}.

A collision forest is a family of subsets of slots, any two disjoint or nested. The contact construction subtracts normal Taylor jets along their collision diagonals. The next result concerns this fixed subtraction, with J.2.3's local order bounds.

Theorem 7.7 (Contact extension)**.** *For every state ζ, arity n and list $\boldsymbol{A} \in \mathbb{F}^n$, the contact construction of Section 18, J.2 (Definitions D36–D38), fixed on the underlying prescription of J.1.1, gives a distribution $\mathfrak{D}^{\zeta}_{\boldsymbol{A}}$ on $\mathbb{R}^{4n}$ such that, for every separated test f,*

$$\mathfrak{D}^{\zeta}_{\boldsymbol{A}}(f) = \int f(x) S^{\zeta}_n(\boldsymbol{A}; x)\, dx\,.$$

The proof is in Section 18, J.2.1–J.2.3. For separated f, all its jets on collision diagonals vanish, so the subtractions leave the correlation integral unchanged. For general f, the contact projections retain compatible field components and the lattice shell bounds make the subtracted sums converge to a continuous functional. J.2 specifies the real stratum bases, scales and local order bounds, independently of OP's estimates. CT enters Theorem B; reconstruction and Theorem A use separated correlations.

8 Main theorems

The existence theorem concerns one Wilson family $\mathcal{W}$, chosen before any prescribed volume sequence. Its tokens $v \in \mathcal{T}$ label the lattice insertions. We first specify its continuum conclusion; the field calculus and stress tensor established above are additional properties of the construction.

Definition 8.1 (Q0: Hierarchy data)**.** Hierarchy data over $\mathcal{W}$ consist of:

- complex Hilbert and vector spaces $\mathcal{H}, \mathcal{L}$, and the relativistic data of Definitions 6.3–6.4;
- finite-dimensional complex normed spaces F_d, linear maps $\iota_d : F_d \to \mathcal{L}$, and $\pi_{\mathrm{quo}} : \mathcal{T} \to \mathbb{F}$, where $\mathbb{F} = \bigsqcup_d F_d$;
- a compact metric space X, a Borel measure ν and $\xi_* \in \mathsf{X}$;
- functions $S^{\zeta}_n : \mathbb{F}^n \times (\mathbb{R}^4)^n \to \mathbb{C}$ for $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$, and $\mathsf{F}_n : \mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n \to \mathbb{C}$;
- finite sets Ch_j, fields $O_{j,\lambda} \in F_j$, and coefficients $c_{j,\lambda} : \mathbb{F}^{I_{\mathrm{cl}}} \times (\mathbb{R}^4)^{I_{\mathrm{cl}}} \times \mathbb{R}^4 \to \mathbb{C}$ for every finite index set I_{cl};
- an element $\Theta \in F_4$.

No further condition is part of Q0. Put $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$. Here the spaces and maps are arbitrary: no response realization, stress interpretation, multilinearity or channel basis property is implicit. Correlations are defined at all positions; the axioms constrain their separated values.

In Q2–Q6, all cited definitions and formulas are read with these hierarchy data. At ξ_*, identify $\Omega_{\xi_*} = \Omega$, $\Phi_{\xi_*} = \Phi$ and $\mathsf{F}_n^{\xi_*} = \mathsf{F}_n$.

Definition 8.2 (Q1: Regular Wilson family)**.** A Wilson family is regular if its insertions are continuous and gauge invariant, each token v has a finite stencil $\Sigma_v \subset \mathbb{Z}^4$ valid for every N, p, y, and some $b_* > 0$ satisfies

$$\forall \varepsilon > 0\ \exists N_0\ \forall N \geq N_0\ \forall p : \quad \left| \frac{\beta_{N,p}}{\log(1/a_N)} - b_* \right| < \varepsilon \,.$$

Only links with both ends in $y + \Sigma_v$ modulo the torus are read.

The relativistic axioms Q2 are Definition 6.9. The following conditions specify the labels, phase average and selected continuation.

Definition 8.3 (Q3: Labels and phases)**.** Every nonzero $\ell \in \mathcal{L}$ has $\Phi(\ell, f) \neq 0$ for some $f \in \mathcal{S}$. Each ι_d is injective, their images cover $\mathcal{L}$, π_{quo} is onto, and $\Phi(\iota(A), f) \neq 0$ for some $A \in \mathbb{F}$, $f \in \mathcal{S}$. The measure ν is a probability measure of full support. Each S_n^ζ is simultaneously permutation symmetric and smooth on Conf_n. For every n, $\boldsymbol{A} \in \mathbb{F}^n$ and $x \in \mathrm{Conf}_n$,

$$\xi \longmapsto S_n^\xi(\boldsymbol{A}; x) \text{ is continuous}, \qquad S_n^{\mathrm{mix}}(\boldsymbol{A}; x) = \int_{\mathsf{X}} S_n^\xi(\boldsymbol{A}; x)\, d\nu(\xi) \,.$$

Definition 8.4 (Q4: Selected continuation)**.** The functions F_n satisfy the ordered-time continuation condition of Definition 6.14 and the boundary condition of Definition 6.13 at ξ_*. In those definitions use the relativistic datum and the maps ι_d of the hierarchy data, and write $\mathsf{F}_n^{\xi_*} = \mathsf{F}_n$, $\Omega_{\xi_*} = \Omega$, $\Phi_{\xi_*} = \Phi$.

Definition 8.5 (Q5: Products)**.** For the channels, fields, coefficients and tables S^ζ of Q0, form $\mathcal{R}_D^\zeta$ by Definition 7.5. These coefficients and remainders satisfy the two inequalities of Theorem 7.6: the coefficient threshold depends on $I_{\mathrm{cl}}, \boldsymbol{A}, M_{\mathrm{geo}}, \eta, \varepsilon, j, \lambda$; the remainder threshold depends on $I_{\mathrm{cl}}, I_{\mathrm{sp}}, \boldsymbol{A}, \boldsymbol{B}, M_{\mathrm{geo}}, \eta, \delta_{\mathrm{sp}}, \varepsilon, D$. Both precede positions and states. Coefficients have no spectator, state or truncation argument.

Definition 8.6 (Q6: Trace test)**.** For the chosen $\Theta \in F_4$, put

$$G_\Theta(r) = S_2^{\xi_*}(\Theta, \Theta; -re_4/2, re_4/2), \qquad A_G = \frac{3 d_{\mathfrak{g}}}{\pi^4} \,.$$

For every $f \in \mathcal{S}$, require

$$\langle \Omega, \Phi(\iota_4 \Theta, f)\Omega \rangle = 0, \qquad \lim_{r \downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0 \quad \text{in } \mathbb{C} \,,$$

and require the finite-profile exclusion of Proposition 7.4.

For the lattice limit, $S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ denotes the kernels

$$K_n(v; x) = S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}}(v_1), \dots, \pi_{\mathrm{quo}}(v_n); x) \,.$$

Q2–Q4 describe a relativistic theory continued from one phase; Q5 controls products, and Q6's inverse logarithm excludes every finite Wick profile. In Theorem A, Q6 places a condition on a degree-four field without requiring a stress-tensor interpretation.

Theorem A (Existence)**.** *For every gauge group of Definition 2.1, there is a regular* (Q1) *Wilson family* $\mathcal{W}$ *such that, for every strictly increasing* $\mathsf{p} : \mathbb{N} \to \mathbb{N}$*, there are a strictly increasing* q *and hierarchy data over* $\mathcal{W}$ (Q0) *satisfying* Q2–Q6*, with*

$$\mathcal{W} \longrightarrow S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}} \quad \textit{along } \mathsf{p} \circ \mathsf{q} \quad \textit{in the sense of Definition 4.3}.$$

Step 10 below deduces Theorem A from Theorem B.

In the construction theorem, the constructed hierarchy means the following Q0 data. Its spaces $F_d = V_d/\mathcal{N}_d(S)$, response labels $\mathcal{L}$ and maps ι_d are those of Definition 4.8, and

$$\pi_{\mathrm{quo}}(d, \mathbf{c}) = (d, [\mathbf{c}]) .$$

Its mixture and phase tables are the quotient correlations of S and Theorem 6.2 on Conf_n, extended arbitrarily on collision diagonals. Its phase space, measure and selected point are those of that phase decomposition, with ξ_* chosen as in H.1.3. Its relativistic data and continuation F_n come from Theorem 6.15 at ξ_*. Its Θ is the centered trace of the tensor in Theorem 5.4, as in Definition 7.1; its channels, fields and coefficients are those constructed in Section 18, J.1.1.

The conclusions B1–B6 retain the following properties of these objects. The tags name the preceding results; the letter-and-number references in Theorem B specify the additional conclusions in Part II.

B1. The kernel, spacings, divisible periods, Taylor and reference normalizations and specified recursion are those fixed above. MT (Theorem 3.6) gives the trajectories and uniform running slope. Endpoint tuning in E.1.3 gives a width $w_{\mathrm{tun}} > 0$, independent of N, p: the bare coupling reaching $\bar{\beta} + w_{\mathrm{tun}} s$ is unique within the specified recursion for $-1 \le s \le 1$ and extends holomorphically to $|s| < 1$.

B2. UV (Theorem 4.2) and E.3.1–E.3.2 give the full fixed-volume limit, including derivatives, interpolation at moving positions and smearing.

B3. IV (Theorem 4.7) and FC (Proposition 4.10) give the quotient calculus and Maxwell dimensions. MC (Proposition 5.2) and SC (Proposition 5.5) give the ordinary and differentiated comparisons. ST (Theorem 5.4) and PH (Theorem 6.2) give the mixed and phase Euclidean, rotation and stress laws, with Θ identified as above.

B4. RC (Theorem 6.15) reconstructs every phase, with the OS comparison maps, ground-space and weak-commutant conclusions. H.2.1–H.2.4 also give low-energy bounds, continuation of the stress tensor and phase averages of joint distributions.

B5. TR (Theorem 7.2) and NF (Proposition 7.4) give the full-radius law and finite-profile exclusion. Sections 17.1 and 17.2 also give the centering identity, unique spectral measure, high-energy limits and polynomial mass-shell and Fock countertests.

B6. OP (Theorem 7.6) and J.1.1–J.1.6 give one product prescription with separated, integrated and nested estimates. CT (Theorem 7.7) and J.2.1–J.2.3 give contact extensions, real bases and subtraction scales. Both use the same original fields and correlation functionals, with separate contact projections, separating functionals and clearance.

The next theorem realizes Theorem A's hierarchy by these quotient fields, with Θ the centered trace of a normalized stress tensor and with every phase reconstructed.

Theorem B (Construction)**.** *For every gauge group of Definition 2.1 and* $\mathsf{b} > 0$*, there is one regular Wilson family* $\mathcal{W}$*, given by Definitions 3.1 and 3.8, satisfying* MT, E.1.1–E.1.3, A.1, E.2.1–E.2.1a (B1)*, and* UV, E.3.1–E.3.2 (B2)*. For every strictly increasing* p*, there are a strictly increasing* q*, the table* S *of* IV *and the constructed hierarchy over* $\mathcal{W}$*, formed from these same objects, satisfying* Q2–Q6*, with* $\mathcal{W} \to S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ *along* $\mathsf{p} \circ \mathsf{q}$ (Definition 4.3)*. Its table, fields and tensor obey* IV, FC, F.2–F.3, MC, ST, SC, PH, G.2.1–G.2.3, G.3.1–G.3.3, H.1.3 (B3)*; its phase theories obey* RC, H.2.1–H.2.4 (B4)*; its* Θ *obeys* TR, NF, I.1.1–I.1.3, I.2.1–I.2.4 (B5)*; its product and contact constructions obey* OP, J.1.1–J.1.6, CT, J.2.1–J.2.3 (B6)*, with the same underlying prescription.*

Steps 1–9 below prove Theorem B. Additional position-space normalization results for degree-four fields are given with I.1.3, and estimates along general collision subspaces with J.2.3; these are separate from the conclusions above. Restricting the full ultraviolet limit also gives convergence along every spacing subsequence.

Deduction of Theorems A and B.

1. Fix G and $\mathsf{b} > 0$. Appendix A and Sections 9–13 supply the common choices, admissible classes, recursion and later lattice estimates. Wilson initialization D.3 starts the induction; Theorem 3.5 advances it; Theorem 3.6 and E.1.3 tune trajectories to the common endpoint. These choices precede any prescribed volume sequence.

2. Fix the local sections, reference levels and terminal maps on those trajectories. Definition 3.8 defines every insertion, including below its reference level. The gauge-invariant local sections are polynomial in the link entries on their stencils; the terminal coefficient maps preserve continuity, gauge invariance and tokenwise locality. Taking $m = N$ in Theorem 3.6 gives Q1 with $b_* = \kappa_G > 0$. This fixes the regular Wilson family.

3. Proposition 4.1 and fixed-volume comparison with analytic tuning give Theorem 4.2 and E.3.1–E.3.2, for every volume and fixed field/difference request. Thus each admissible test has a full inner limit before volume extraction.

4. Given any strictly increasing p, Theorem 4.7 chooses q, S, including the refinement for corrected coefficient maps. In Section 14, F.1 identifies the extracted volume kernels with the full kernels by uniqueness of limits. Since $L_{\mathsf{p}(\mathsf{q}(k))} \to \infty$, every compact support eventually fits. This proves the approximation of Definition 4.3, without a spacing refinement.

5. Define $F_d = V_d/\mathcal{N}_d(S)$ and $\mathcal{L}$ by Definition 4.8. Proposition 4.10 and F.2 give inclusions, derivatives, dimensions and the response realization. Proposition 5.2 and the current lemmas of Section 15 give the conserved line; Theorem 5.4 gives its normalized tensor, and the rotations are those of Section 15; Proposition 5.5 gives the differentiated comparison. The volume refinement is unchanged.

6. Theorem 6.2 and H.1.1–H.1.3 of Section 16 construct X, ν, S^{ξ} on the same quotient spaces. Choose ξ_* from the full-measure set with one-dimensional time-axis OS ground space. This selection precedes reconstruction and uses only that ground-space property. Theorem 6.15 reconstructs every character simultaneously; H.2.1–H.2.4 retain this selected theory and give B4.

7. Set $\pi_{\mathrm{quo}}(d, \mathbf{c}) = (d, [\mathbf{c}])$. Representatives make this map onto. Every response lies in one $\iota_d F_d$, and each ι_d is injective. The unit response is nonzero since its one-point function is one; faithfulness gives a nonzero acting field. Extend the quotient mixture and phase correlations arbitrarily off Conf_n. Separated tests vanish on collision diagonals, so these extensions do not affect Definition 4.3. Faithfulness in Theorem 6.15 and the preceding facts give the label

clauses of Q3; E1, E6 and Theorem 6.2 give its phase clauses; reconstruction gives Q2 and Q4. These identifications use the all-spectator quotient.

8. Center the trace of this tensor in this selected phase. Theorem 7.2 gives Q6's vacuum-centering, amplitude and full-radius clauses; I.1.1–I.1.3 of Section 17 give the operator identity and the unique spectral measure. Proposition 7.4 gives Q6's finite-profile exclusion; I.2.2–I.2.4 give the additional countertests in B5, with ξ_* unchanged.

9. Choose the prescription of J.1.1 once. Theorem 7.6 gives Q5; J.1.3–J.1.6 give the integrated and nested bounds in B6. Theorem 7.7 and J.2.1–J.2.3 of Section 18 use the same underlying prescription, with separate contact projections, functionals and clearance, to give the extensions, real bases and subtraction scales. This uses its own lattice estimates.

10. Steps 1–9 give B1–B6 on the same family, table, quotient fields, tensor, phases and selected theory. Retain Q1–Q6 and set $\mathsf{b} = 1$. Their quantifier order, lattice approximation, reconstruction, products and trace test give Theorem A.

□

Part II
Proofs

9 Fields, labels and norms

This section is under construction.

10 The renormalization step

This section is under construction.

Definition 10.1 (Axial frames)**.** This definition will be stated here; the section is under construction.

Definition 10.2 (Coarse predictors)**.** This definition will be stated here; the section is under construction.

11 Effective actions and the marginal flow

This section is under construction.

Definition 11.1 (Curvature slots)**.** This definition will be stated here; the section is under construction.

Definition 11.2 (Path action)**.** This definition will be stated here; the section is under construction.

12 The nonlinear step

This section is under construction.

Definition 12.1 (Regular activities)**.** This definition will be stated here; the section is under construction.

Definition 12.2 (Forced activities)**.** This definition will be stated here; the section is under construction.

Definition 12.3 (Continuity of representations)**.** This definition will be stated here; the section is under construction.

13 The multiscale construction

This section is under construction.

14 Infinite volume

This section is under construction.

15 The Maxwell comparison, Ward identities and the stress tensor

This section is under construction.

16 Phases and reconstruction

This section is under construction.

17 The trace law and non-freeness

This section is under construction.

18 The operator product expansion and contact extensions

This section is under construction.

A Constants and order of choice

Table 1 gives the choices for the construction. At each fixed stage the requirements in Sections 10–13, 15.2, 17.1 and 18 impose finitely many lower bounds on a ratio or an entry level, depending only on earlier data. Costs determining a ratio precede that ratio; increasing an entry level preserves earlier requirements. The displayed inequalities are properties of these choices. They do not define a class of constructions for which the conclusions are asserted.

Table 1: Choices in dependence order. Stage choices are made at each fixed $j \geq 4$ before entry levels.

	Choice	Dependence, restrictions and purpose
1	Gauge normalization; $\mathsf{b} > 0$	Fix G, its realization, probability Haar measure and metric, then $d_{\mathfrak{g}}, C_{\mathfrak{g}}, b_0, \kappa_G$ as in Definition 2.1; $\kappa_G > 0$. All later choices use these data.
2	Frame and predictor cutoffs; axial factor	Depend on G. The frame and predictor cutoffs select the unique-minimizer branches of the block kernel. On a radius-R patch, the total link chord in an eligible axial frame is at most $a_{\mathrm{ax}}(2R+1)$ times the total plaquette chord, with $a_{\mathrm{ax}} \geq 16$. Choose the cutoffs small enough for branch validity on every patch (Section 10.1).
3	Homogeneous bases, norms, sections, stencils	Depend on G, d. Impose Taylor matching, reality and H_4 covariance (Section 9.1 and Definitions 3.7–3.8).
4	Stage parameters	Depend on rows 1–3 and j. Choose derivative orders $\mathfrak{q}_j \geq 6$, spatial powers $\varpi_j \geq 2$ and support exponents $\sigma_j = 100$. In Section 11, C.1, $w_j^{\mathrm{path}} > 0$ multiplies path length in the exponential weight, and $A_j^{\mathrm{path}} > 0$ bounds the weighted row, cubic and quartic sums. The constant $C_j \geq 1$ bounds marked coefficient transport (E.2.2). Clipping coefficients satisfy $c_{\mathrm{dir},j} > 16 c_{\mathrm{lin},j} > 0$; they set the quadratic and linear pieces of the clipped curvature action. Its health cutoff relaxes with distance through $e^{\mathrm{dist}/R_j^{\mathrm{rel}}}$, where $R_j^{\mathrm{rel}} > 0$ and $R_j^{\mathrm{rel}} w_j^{\mathrm{path}} \geq 12$. Choose $\omega_j^{\mathrm{RG}} \geq 0$ to bound the residual in Theorem 3.5's coupling formula at stage j. Orders and powers tend to infinity and, along levels, are nondecreasing, as are support exponents. All ratio costs are now fixed.
5	Odd ratios ℓ_j^{st}	Depend on rows 1–4; choose $\ell_j^{\mathrm{st}} \geq 3$ nondecreasing and above every ratio threshold. At $j = J_{m+1}$ and $\ell_m = \ell_j^{\mathrm{st}}$ require $$\frac{\omega_j^{\mathrm{RG}}}{\log \ell_m} \leq \min(\kappa_G/4, j^{-2}), \qquad \frac{\log(1 + A_j^{\mathrm{path}} + (w_j^{\mathrm{path}})^{-1} + C_j + R_j^{\mathrm{rel}} + \sigma_j)}{\log \ell_m} \leq j^{-2}.$$ These bounds control the drift and logarithmic stage costs relative to $\log \ell_m$.
6	Exponents and coupling entry bounds	Depend on rows 1–5. The positive antitone exponents θ_j control the quadratic loop bound in row 9. The bounds β_j^{ent} are lower bounds on the reference coupling when stage j is used. Choose them to meet the finite maximum of the remaining stage requirements, including those of Sections 15.2, 17.1 and 18. Entry bounds may be raised.

Table 1 (continued)

	Choice	Dependence, restrictions and purpose
7	$\bar\beta$, entry levels m_j^{ent}, minimum periods $n_j^{\min}$	Depend on rows 1–6. The levels m_j^{ent} delay the first use of stage j; $n_j^{\min}$ are its minimum lattice periods. Choose $\bar\beta > 1$ large enough for the initial coupling requirements and for $t_m \le 1$ at every level. Choose entry levels and periods to satisfy the remaining level and geometric requirements. Delaying entry preserves earlier requirements.
8	$J_m, \ell_m, D_m, a_m, L_p$, $m_{\mathrm{ref}}(d)$	Depend on rows 1–7. Set $D_0 = 1$, $D_{m+1} = \ell_m D_m$, $a_m = D_m^{-1}$, $s_m = \log D_m$, with $\ell_m = \ell^{\mathrm{st}}_{J_{m+1}}$ and $4 \le J_m \le J_{m+1} \le J_m + 1$, $J_m \to \infty$. Require $0 < \theta_{J_m} \le J_m^{-1}$, $\log \ell_m \to \infty$, $\log \ell_m / s_m \to 0$. Choose strictly increasing L_p with $4 \mid L_p$ and every positive integer dividing infinitely many L_p. Require $L_p D_m \ge 100(B_m + 1)$ and $L_p D_{m+1} \ge 100((J_{m+1}+3)\ell_m + 1)$. Choose $J_{m_{\mathrm{ref}}(d)} \ge d + 3$, with $\mathfrak{q}_{J_m} \ge d+1$, $\varpi_{J_m} \ge d$ for $m \ge m_{\mathrm{ref}}(d)$.
9	Derived level scales	No new choices. Set $$\widehat\beta_m = \bar\beta + \kappa_G s_m > 1, \qquad M_m^{\mathrm{cut}} = \exp((\log\log\widehat\beta_m)^2)\,,$$ $$t_m = \widehat\beta_m^{-1/2} M_m^{\mathrm{cut}}, \qquad B_m = \lceil (\log\widehat\beta_m)^2 \rceil\,,$$ $$\epsilon_m^{\mathrm{reg}} = \widehat\beta_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}}\,,$$ $$\Lambda_m^{\mathrm{loop}} = 2(M_m^{\mathrm{cut}})^{\theta_{J_m}}, \qquad \mathfrak{B}_m = [\widehat\beta_m/2, 2\widehat\beta_m]\,.$$ These give the field thresholds, activity bounds and coupling bands. The quantity $\Lambda_m^{\mathrm{loop}}$ bounds the weighted quadratic loop sum in Section 11, C.1.
10	Actions, classes, recursion, trajectories, couplings; width w_{tun}	Depend on rows 1–9. Choose the path actions and admissible subsets, the specified recursion and its trajectories for every finite N, p, all with the common level-zero endpoint (Theorem 3.6). In Section 13, E.1.3, $w_{\mathrm{tun}} > 0$ is the half-width of the endpoint interval $\bar\beta + [-w_{\mathrm{tun}}, w_{\mathrm{tun}}]$, independent of N, p. The tuned bare coupling for endpoint $\bar\beta + w_{\mathrm{tun}} s$ extends holomorphically in s to $\lvert s\rvert < 1$.
11	Marked transfers, terminal maps, insertions	Depend on rows 1–10; impose reference normalization (Definition 3.8). Fix them before any prescribed volume sequence.
12	$\mathfrak{p}$, then $\mathfrak{q}, S, F_d, \partial_a, \mathcal{L}$	Prescribe arbitrary increasing $\mathfrak{p}$ first. Theorem 4.7 and Proposition 4.10 give the refinement and continuum fields. Theorem 4.2 retains the full inner limit.
13	Comparison radii, frames, J, charges, $c_\star, R, T$	Depend on row 12 and the fixed lattice family. Use Proposition 5.2, then Theorem 5.4 and Proposition 5.5. Refine only comparison radii.
14	Slab operators, X, ν, S^ξ, then ξ_*	Depend on earlier data. Theorem 6.2 gives the phase decomposition. Choose ξ_* from the full-measure set with one-dimensional time-axis OS ground space (H.1.3). This choice precedes reconstruction and uses only that ground-space property.

Table 1 (continued)

	Choice	Dependence, restrictions and purpose
15	Every-phase theories and continuations; Θ, μ_Θ	Depend on earlier data. Theorem 6.15 reconstructs the theories. Definition 7.1 centers the trace in the already selected phase; Theorem 7.2 gives its short-distance law; I.1.2 supplies its spectral measure.
16	OPE prescription; contact projections, distributions, bases, scales	Depend on rows 12–14 and the fixed family. Theorems 7.6 and 7.7 use the same original fields and correlation functionals. The contact projections, separating functionals and clearance are chosen separately from those used for the product coefficients.
17	Estimate constants and thresholds	Depend only on the fixed data preceding them in each statement. They have no later position, volume or state dependence wherever uniformity is asserted.

References